Scenario: “BrightPath Solutions’ Recruitment Process”

BrightPath Solutions is a **growing technology company** that specializes in developing software and cloud-based solutions for clients in various industries. As the company expands, it needs to hire more qualified staff to meet increasing project demands.

To improve efficiency and consistency in hiring, the **Human Resources (HR) Department** wants to design a **Business Process Model and Notation (BPMN) diagram** that captures the entire **Recruitment Process** — from the approval of a hiring request to the successful onboarding of a new employee.

You have been assigned as the **Business Process Analyst** responsible for documenting and modeling this process.

Process Description

1. The process begins when a **Hiring Request Form** is **approved** by a **Department Manager**. This approval serves as the **event trigger** that initiates the recruitment process.
2. Once the request is received, the **HR Officer** reviews the details of the job vacancy, verifies the budget, and confirms the job requirements.
3. After verification, the **HR Assistant** prepares and **posts the job advertisement** on the company website and external job portals.
4. Interested applicants submit their **resumes and application letters** through the company’s online recruitment system.
5. The **system automatically receives and stores applications**, after which the **HR Officer** screens the resumes based on qualifications and experience.
6. Qualified applicants are shortlisted, and the **HR Assistant** contacts them through **email** to schedule interviews.
7. The **HR Officer** and **Hiring Manager** then **conduct interviews** to assess the candidates’ technical and behavioral competencies.
8. After the interviews, the **Hiring Manager evaluates** the results and recommends the most suitable candidate.
9. The **HR Officer** prepares and **sends a job offer** to the selected applicant via email.
10. The **applicant reviews** the job offer and responds with either **acceptance** or **rejection**.
    - If accepted, the HR Assistant prepares **onboarding documents**, including the employment contract and new hire forms.
    - If declined, the HR Officer updates the recruitment database and proceeds with the next shortlisted candidate.

11. For accepted candidates, the **HR Assistant** simultaneously prepares the **onboarding kit**, updates the **HR system**, and notifies the **IT Department** to set up equipment and system access.
12. The process **ends** when the **candidate is officially hired and onboarded**, or the **offer is declined** and the vacancy remains open for reprocessing.

Business Rules and Policies

1. Only **approved hiring requests** can trigger recruitment.
2. Minimum qualifications must be met before interview scheduling.
3. Each applicant must be evaluated by at least **two interviewers**.
4. Job offers remain valid for **seven (7) calendar days**.
5. All applicant data and hiring documents must be **stored for one (1) year** for compliance.

Student Task

Instructions:

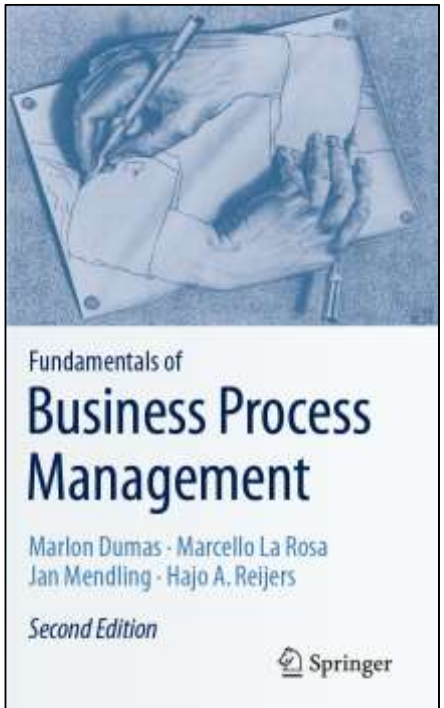
Using the details in the scenario above:

1. Create a **BPMN diagram** of the Recruitment Process.
2. Include the following elements in your diagram:
   - Start and End Events
   - **Activities**
   - **Gateways** for decision points
   - **Message Flows** between participants
   - **Pools / Lanes** showing roles
   - **Artifacts/Business Objects**
   - **Sequence Flows**
3. Name each symbol clearly (Follow naming conventions)

Rubric (100 Points)

Criteria	Description	Points
Correct Use of BPMN Elements	Accurate and consistent use of ALL BPMN Components	30
Completeness	Includes all process components listed (a–h)	25
Logic and Flow	Sequence of activities and decisions is logical	25
Clarity and Labeling	Proper names and clear representation of participants	20
Total		**100**

Chapter 6: Qualitative Process Analysis



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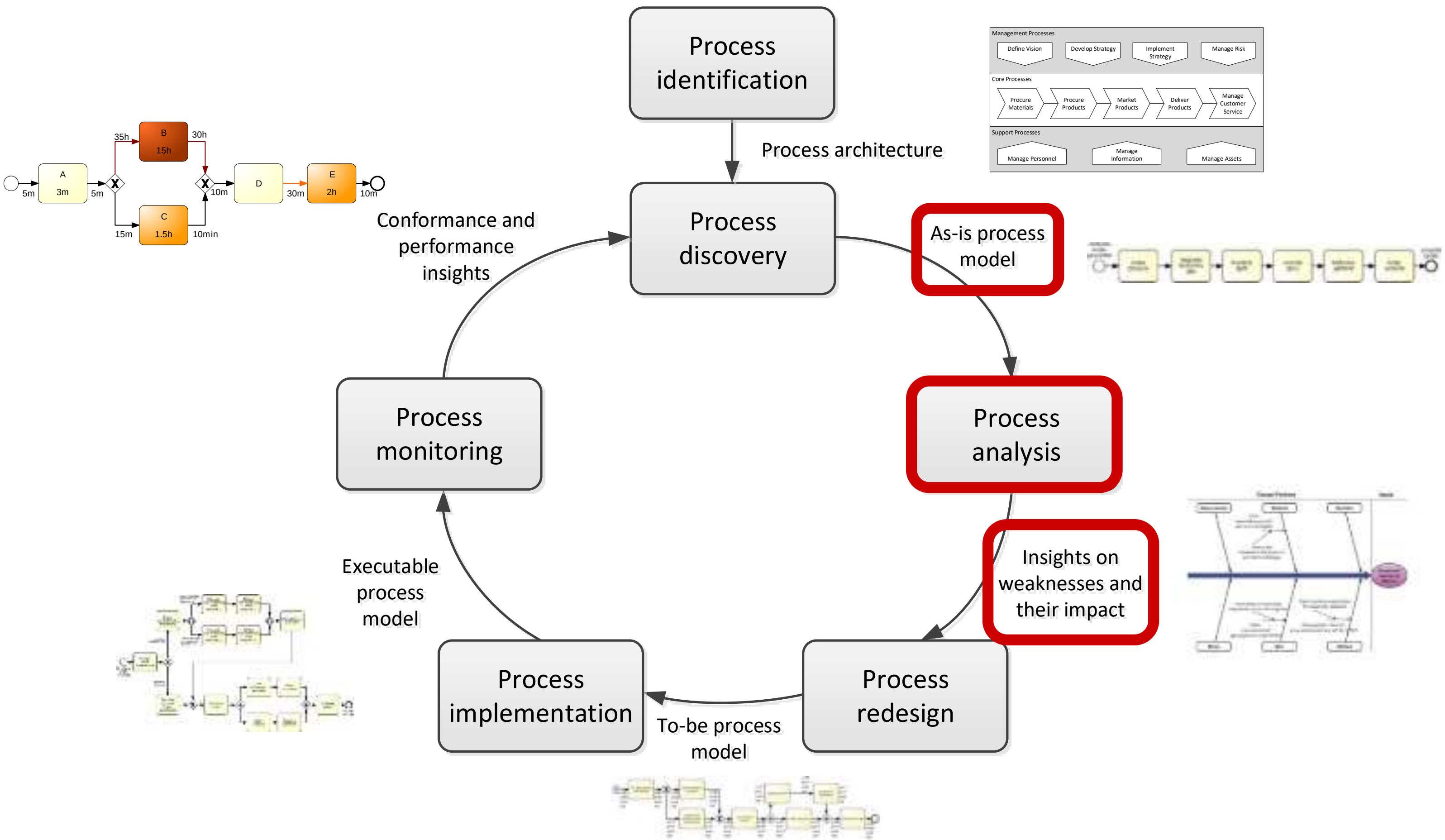
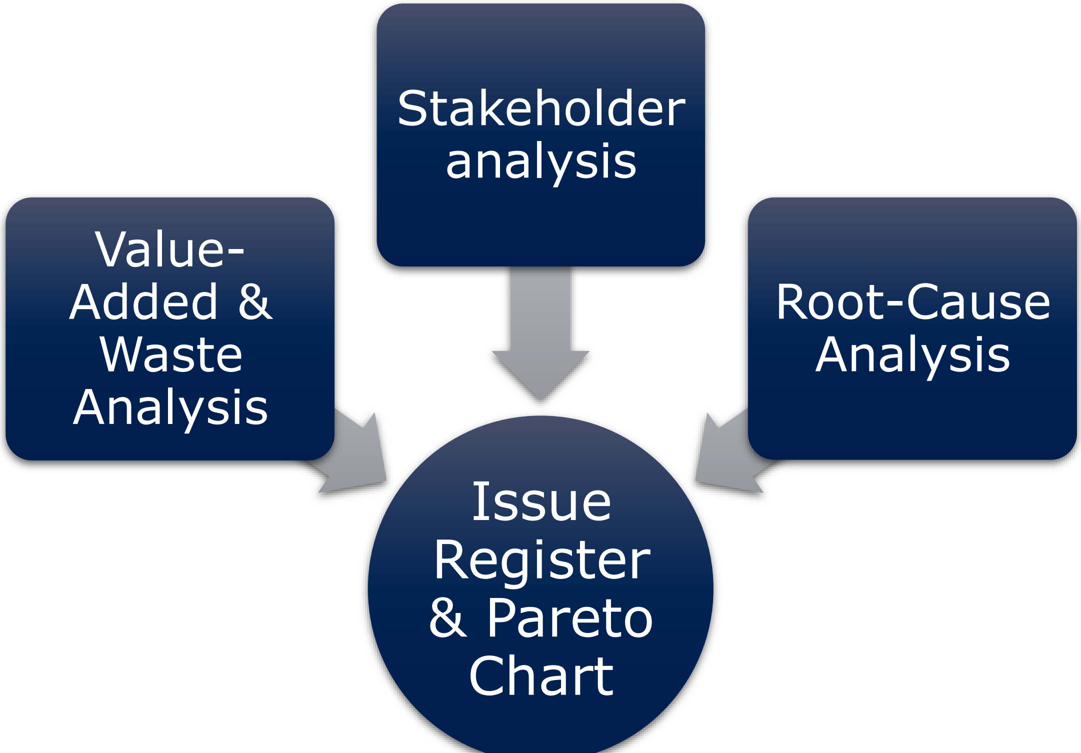
2. Waste Analysis
3. Stakeholder Analysis & Issue Documentation
4. Root-Cause Analysis
5. Recap

Process Analysis in the BPM Lifecycle



Qualitative Process Analysis: Overview



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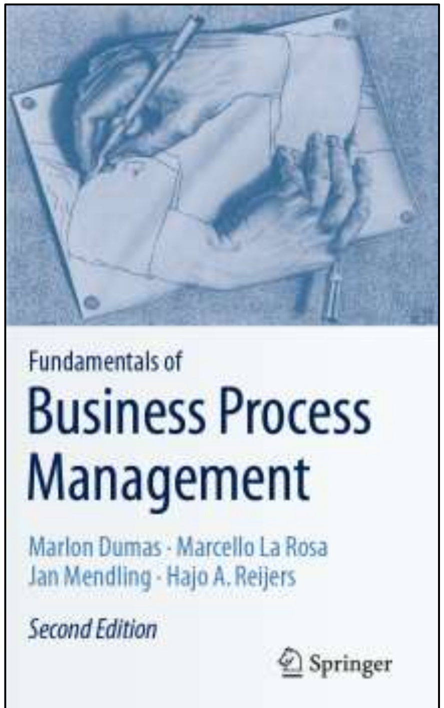

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Value-added analysis

- Decorticate the process into steps
  - Steps performed before a task
  - The task itself, possibly decomposed into smaller steps
  - Steps performed after a task, in preparation for the next task
- Classify each step
  - Value-adding (VA)
  - Business value-adding (BVA)
  - Non-value-adding (NVA)

Value-adding activities

- Produces value or satisfaction to the customer.
- Criteria:
  - Is the customer willing to pay for this step?
  - Would the customer agree that this step is necessary to achieve their goals?
  - If the step is removed, would the customer perceive that the end product or service is less valuable?
- Examples:
  - Order-to-cash process: Confirm delivery date, Deliver products
  - University admission process: Assess application, Notify admission outcome

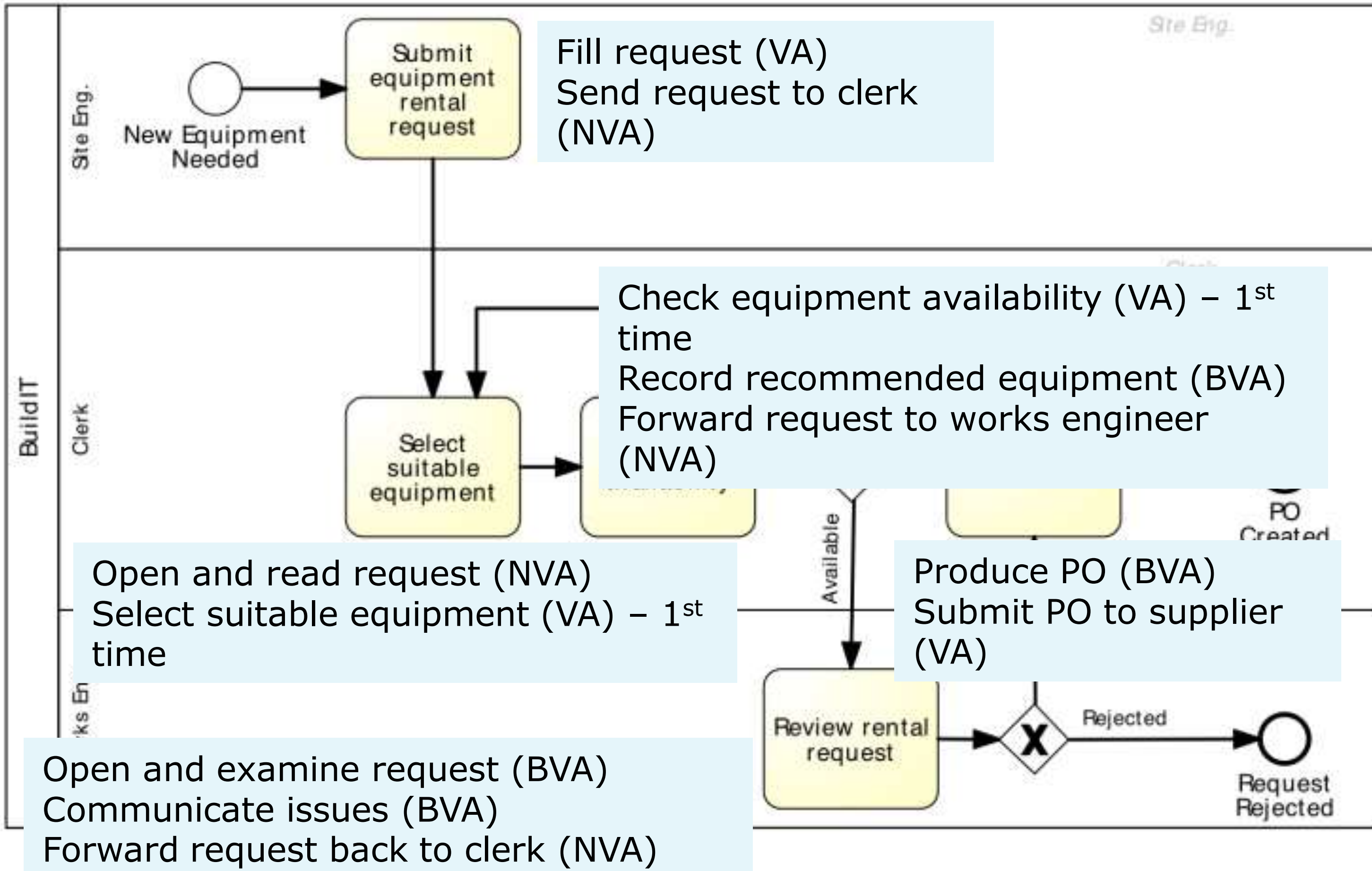
Business value-adding activities

- Necessary or useful for the business to operate.
- Criteria
  - Is this step required in order to collect revenue, to improve or grow the business?
  - Would the business (potentially) suffer in the long-term if this step was removed? Does it reduce risk of business losses?
  - Is this step required in order to comply with regulatory requirements?
- Example
  - Order-to-cash process: Check purchase order, Check customer's credit worthiness, Issue invoice, Collect payment, Collect customer feedback
  - University admission process: Verify completeness of application, Check validity of degrees, Check validity of language test results

Non-value-adding activities

- Everything else besides VA and BVA
- Incudes:
  - Handovers, context switches
  - Waiting times, delays
  - Rework or defect correction
- Examples
  - Order-to-cash: Forward PO to warehouse, Re-send confirmation, Receive rejected products
  - University admission: Forward applications to committee, Receive admission results from committee

Extract of Equipment Rental Process



Fill request (VA)
Send request to clerk (NVA)

Check equipment availability (VA) – 1st time
Record recommended equipment (BVA)
Forward request to works engineer (NVA)

Open and read request (NVA)
Select suitable equipment (VA) – 1st time

Produce PO (BVA)
Submit PO to supplier (VA)

Open and examine request (BVA)
Communicate issues (BVA)
Forward request back to clerk (NVA)

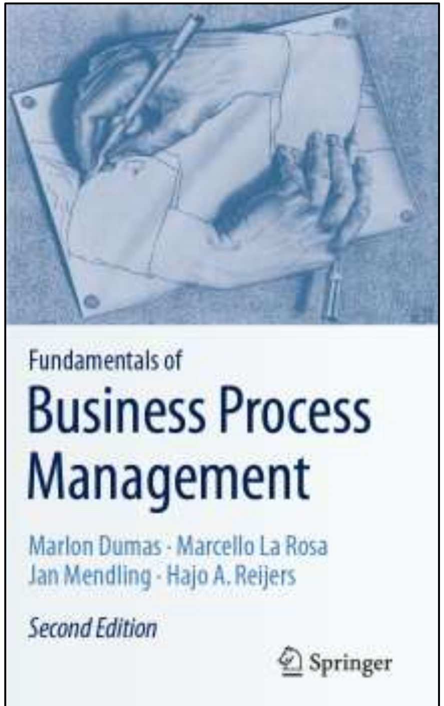
Equipment Rental Process – VA Analysis

Step	Performer	Classification
Fill request	Site engineer	VA
Send request to clerk	Site engineer	NVA
Open and read request	Clerk	NVA
Select suitable equipment	Clerk	VA
Check equipment availability	Clerk	VA
Record recommended equipment & supplier	Clerk	BVA
Forward request to works engineer	Clerk	NVA
Open and examine request	Works engineer	BVA
Communicate issues	Works engineer	BVA
Forward request back to clerk	Works engineer	NVA
Produce PO	Clerk	BVA
Send PO to supplier	Clerk	VA

Exercise

- Pharmacy prescription process (Exercise 1.6)
  - Identify at least three VA steps, at least three BVA steps and at least three NVA steps

Chapter 6: Qualitative Process Analysis



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   2. Hold
   3. Overdo
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Waste analysis

"*All we are doing is looking at the time line, from the moment the customer gives us an order to the point when we collect the cash.*

And we are reducing the time line by reducing the non-value-adding wastes ”

Taiichi Ohno

Seven sources of waste

Move

- Unnecessary Transportation
- Motion

Hold

- Inventory
- Waiting (and idleness)

Over-do

- Defects
- Over-Processing
- Over-Production

Move

Unnecessary transportation

- Send or receive materials or documents (incl. electronic) taken as input or output by the process

Example:

- To apply for admission at a University, students fill in an online form. When a student submits the online form, a PDF document is generated. The student is requested to download it, sign it, and send it by post together with the required documents: 1. Certified copies of degree and academic transcripts. 2. Results of language test. 3. CV.
  When the documents arrive to the admissions office, an officer checks their completeness. If a document is missing, an e-mail is sent to the student. The student has to send the missing documents by e-mail or post depending on document type.

Motion

- Motion of resources internally within the process
- Common in manufacturing processes, less common in business processes

Examples

- Vehicle inspection process, a process worker moves with the inspection forms from one inspection base to another; in some cases inspection equipment also needs to be moved around
- Approval process, a process workers moves around the organization to collect signatures

Hold

Inventory

- Materials inventory
- Work-in-process (WIP)

Examples

- Vehicle inspection process, when a vehicle does not pass the first inspection, it is sent back for adjustments and left in a pending status. At a given point in time, about 100 vehicles are in the “pending” status across all inspection stations
- University admission process: About 3000 applications are handled concurrently

Waiting

- Waiting for materials or input data
- Task waiting for a resource
- Resource waiting for work (resource idleness)

Examples

- Vehicle inspection process: A technician at a base of the inspection station waiting for the next vehicle
- Approval process: Request waiting for approver
- University admission process: Incomplete application waiting for additional documents; batch of applications waiting for committee to meet

Over-do

Defects

- Correcting or compensating for a defect
- Rework loops

Examples

- Vehicle inspection: A vehicle needs to come back to a station due to an omission
- Travel approval: Request sent back to requestor for revision
- University admission: Application sent back to applicant for modification; request needs to be re-assessed later due to incomplete information

Over-processing

- Tasks performed unnecessarily given the outcome of the process
- Unnecessary perfectionism

Examples

- Vehicle inspection: Technicians take time to measure vehicle emissions with higher accuracy than required, only to find that the vehicle clearly does not fulfill the required emission levels
- Travel approval: 10% of approvals are trivially rejected at the end of the process due to lack of budget
- University admission: Officers spend time verifying the authenticity of degrees, transcripts and language test results. In 1% of cases, these verifications uncover issues.
  Verified applications are sent to the admissions committee. The admission committee accepts 20% of the applications it receives

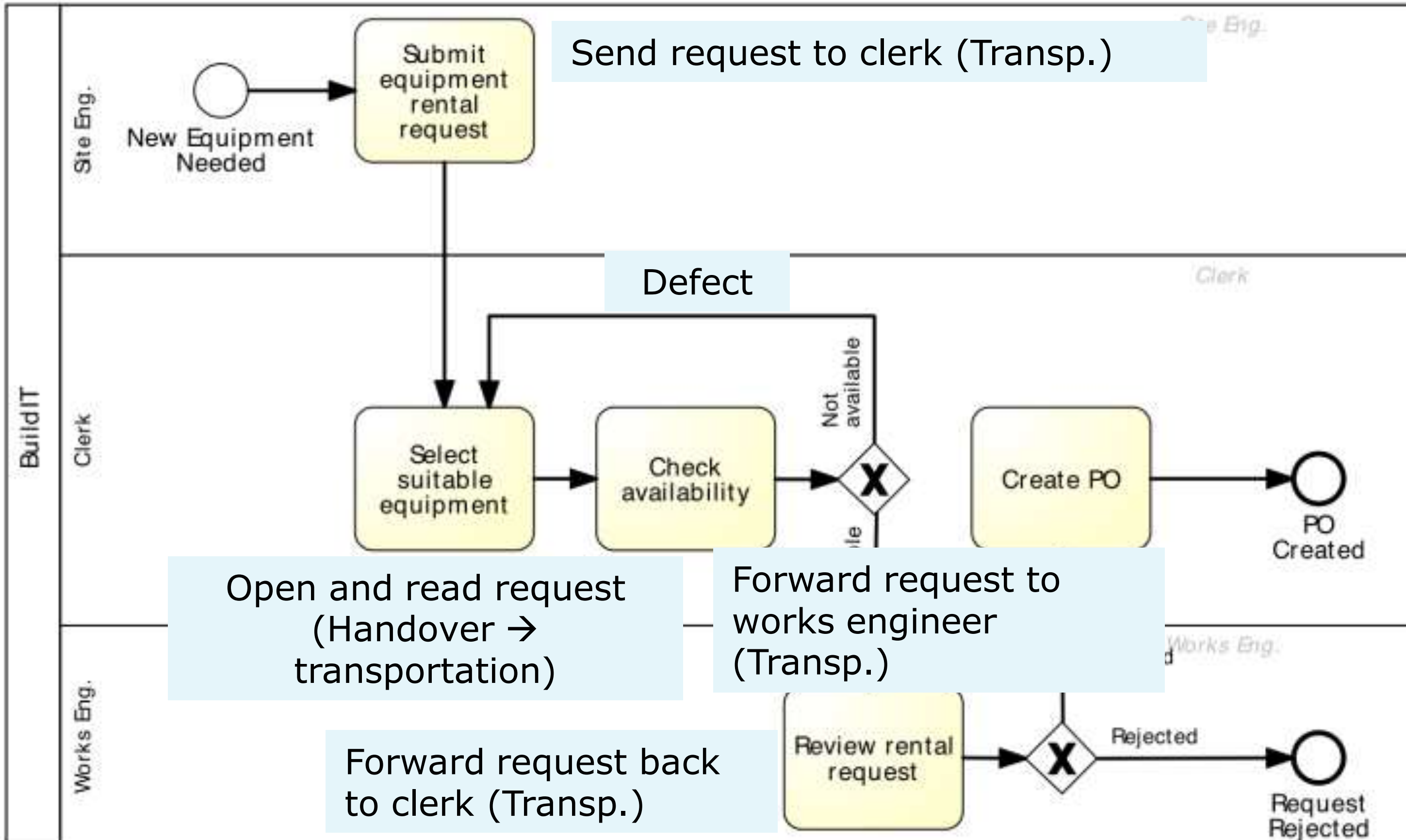
Over-production

- Unnecessary process instances are performed, producing outcomes that do not add value upon completion

Examples

- Order-to-cash: In 50% of cases, issued quotes do not lead to an order
- Travel approval: In 5% of cases, travel requests are approved but the travel is cancelled
- University admission: About 3000 applications are submitted, but only 800 are considered eligible after assessment

Equipment rental process: wastes



Equipment rental process: wastes

Transportation

- Site engineer sends request to clerk
- Clerk forwards to works engineer
- Works engineer send back to clerk

Inventory

- Equipment kept longer than needed

Waiting

- Waiting for availability of works engineer to approve

Equipment rental process: wastes

Defect

- Selected equipment not available, alternative equipment sought
- Incorrect equipment delivered and returned to supplier

Over-processing

- Clerk finds available equipment and rental request is rejected because equipment not needed
- Rental requests being approved and then canceled by site engineer

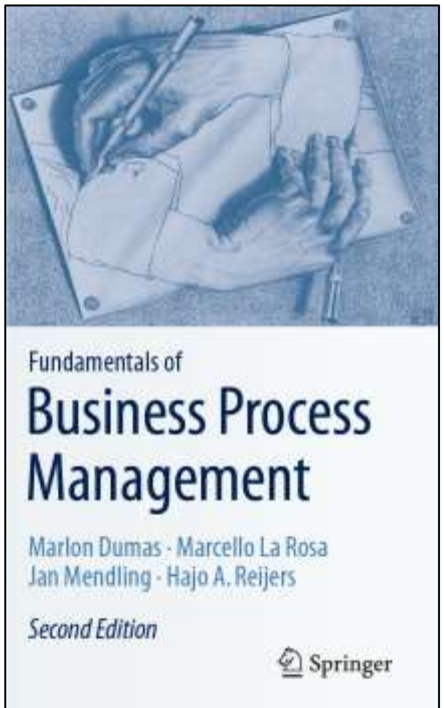
Over-production

- Equipment being rented and not used at all

Exercise

- Pharmacy prescription process (Exercise 1.6)
  - Identify wastes in this process

Chapter 6: Qualitative Process Analysis



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    1. Stakeholder Analysis
    2. Issue Register
    3. Pareto & PICK Chart
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Identifying and documenting process issues

- Stakeholder analysis: allows us to collect issues from complementary perspectives.
- Issue register: allows us to document issues and their impact in a structured manner
- Pareto analysis and PICK charts: allow us to select a subset of issues for further analysis and redesign.

Stakeholder Analysis

Stajeholder analysis is about gathering data from multiple sources by interviewing stakeholders of different types and reconciling their viewpoints

In BPM, stakeholder analysis is commonly used to gather information

about issues that affect the performance of the process from different perspectives.

There are typically five categories of stakeholders:

- The customer(s) of the process.
- The process participants.
- The external parties (e.g., suppliers, sub-contractors) involved in the process.
- The process owner and the operational managers who supervise the process participants.
- The sponsor of the process improvement effort and other executive managers who have a stake in the performance of the process.

Typical Stakeholder Concerns

- Customers are often concerned about slow cycle time, defects, lack of transparency, or lack of traceability (inability to observe the current process status).
- Process participants might be rather concerned about:
  - High resource utilization, working under stress.
  - Defects arising from handoffs in the process and wastes.
- External parties (e.g. suppliers and sub-contractors) are generally concerned about having a steady or growing stream of work from the process, being able to plan their work ahead, and being able to meet contractual requirements.
- The process owner is usually concerned with performance, be it high cycle times or high processing times. Also be concerned about common defects and wastes, and compliance with internal policy and external regulations.
- The sponsor and other high-level managers are generally concerned with the strategic alignment of the process and the contribution of the process to key performance indicators. Also concerned about the ability of the process to adapt to evolving customer expectations, competition, and market conditions.

Issue register

- Purpose: to maintain, organize and prioritize perceived weaknesses of the process (issues)
- Sources of issues:
  - Input to a process modelling project
  - Collected as part of ongoing process improvement actions
  - Collected during process discovery (modelling)
  - Value-added/waste analysis

Issue register structure

- Can take the form of a table with:
  - Issue identifier
  - Short name
  - Description
  - Assumptions
  - Impact: Qualitative and Quantitative
  - Possible improvement actions
- Larger process improvement projects may require issue trackers

Example of an issue documentation

Issue name

- Equipment kept longer than needed

Description

- Site engineers keep rented equipment longer than needed by asking for deadline extensions

Assumptions

- 3000 pieces of equipment rented p.a.
- In 10% of cases, equipment is kept two days more than needed
- Average rental cost is 100 per day

Quantitative impact

- $0.1 \times 3000 \times 2 \times 100 = 60{,}000$ p.a

Issue Register Example

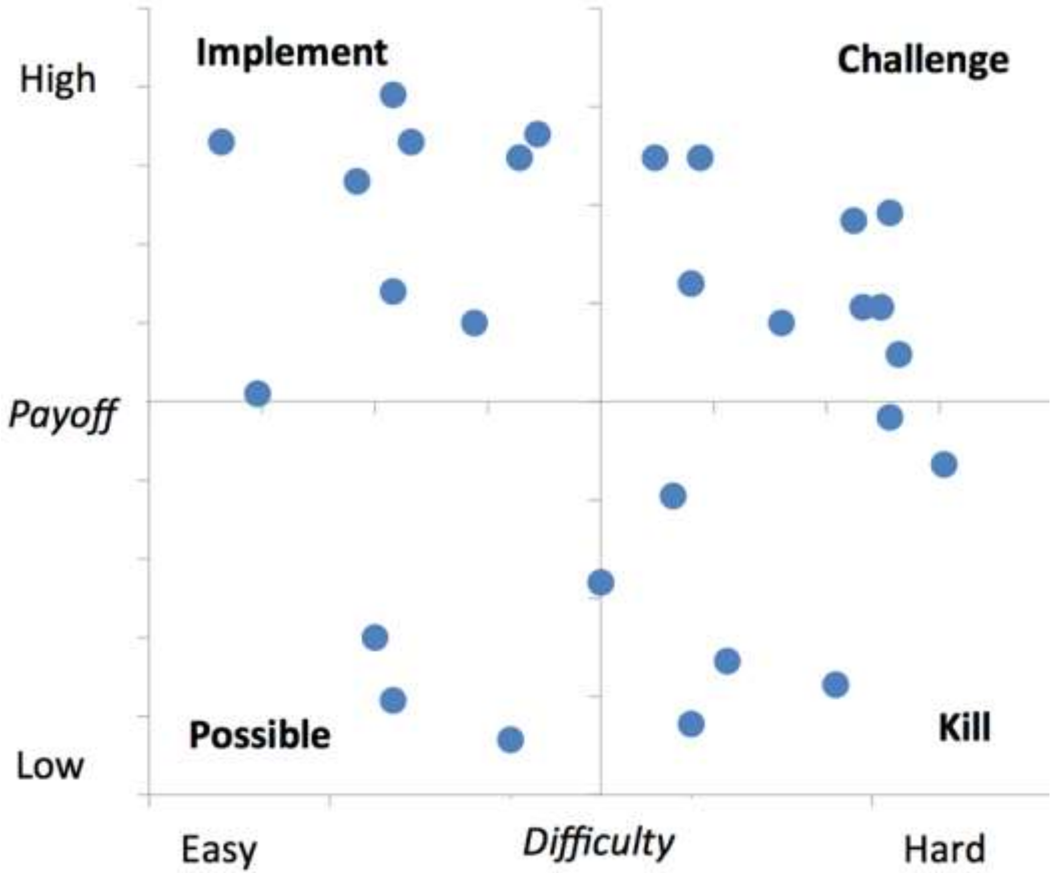
Name	Explanation	Assumptions	Qualitative Impact	Quantitative Impact
Equipment kept longer than needed	Site engineers keep equipment longer than needed via deadline extensions	3000 pieces of equipment rented p.a. In 10% of cases, equipment kept two days longer than needed. Rental cost is 100 per day		$0.1 \times 3000 \times 2 \times 100 = 60{,}000$ p.a.
Rejected equipment	Site engineers reject delivered equipment due to non-conformance to their specifications	3000 pieces of equipment rented p.a. 5% of them are rejected due to an internal mistake For each equipment rejected due to an internal mistake, BuildIT is billed 100.	Disrupted schedules. Employee stress and frustration	$3000 \times 0.05 \times 100 = 15{,}000$ p.a.
Late payment fees	Late payment fees incurred because invoices are not paid by their due date	3000 pieces of equipment rented p.a. Average rental time is 4 days Rental cost is 100 per day. Each rental leads to one invoice. About 10% of invoices are paid late. Penalty for late payment is 2%.	Poor reputation with suppliers	$0.1 \times 3000 \times 4 \times 100 \times 0.02 = 2400$ p.a.

Pareto chart

- Useful to prioritize a collection of issues
- Bar chart where the height of the bar denotes the impact of each issue
- Bars sorted by impact
- Superposed curve of cumulative percentage impact

Pareto chart example

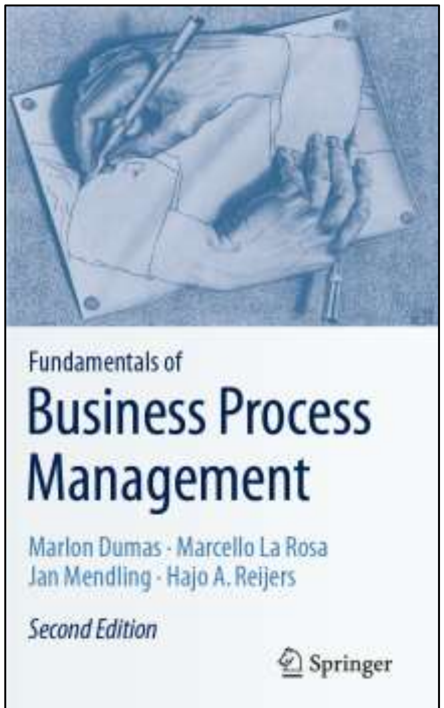
Two-Dimensional Prioritization: PICK Chart



Exercise

- Pharmacy prescription process (Exercise 1.6)
  - Identify and document at least two issues in an issue register

Chapter 6: Qualitative Process Analysis



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    2. Why-Why Diagram
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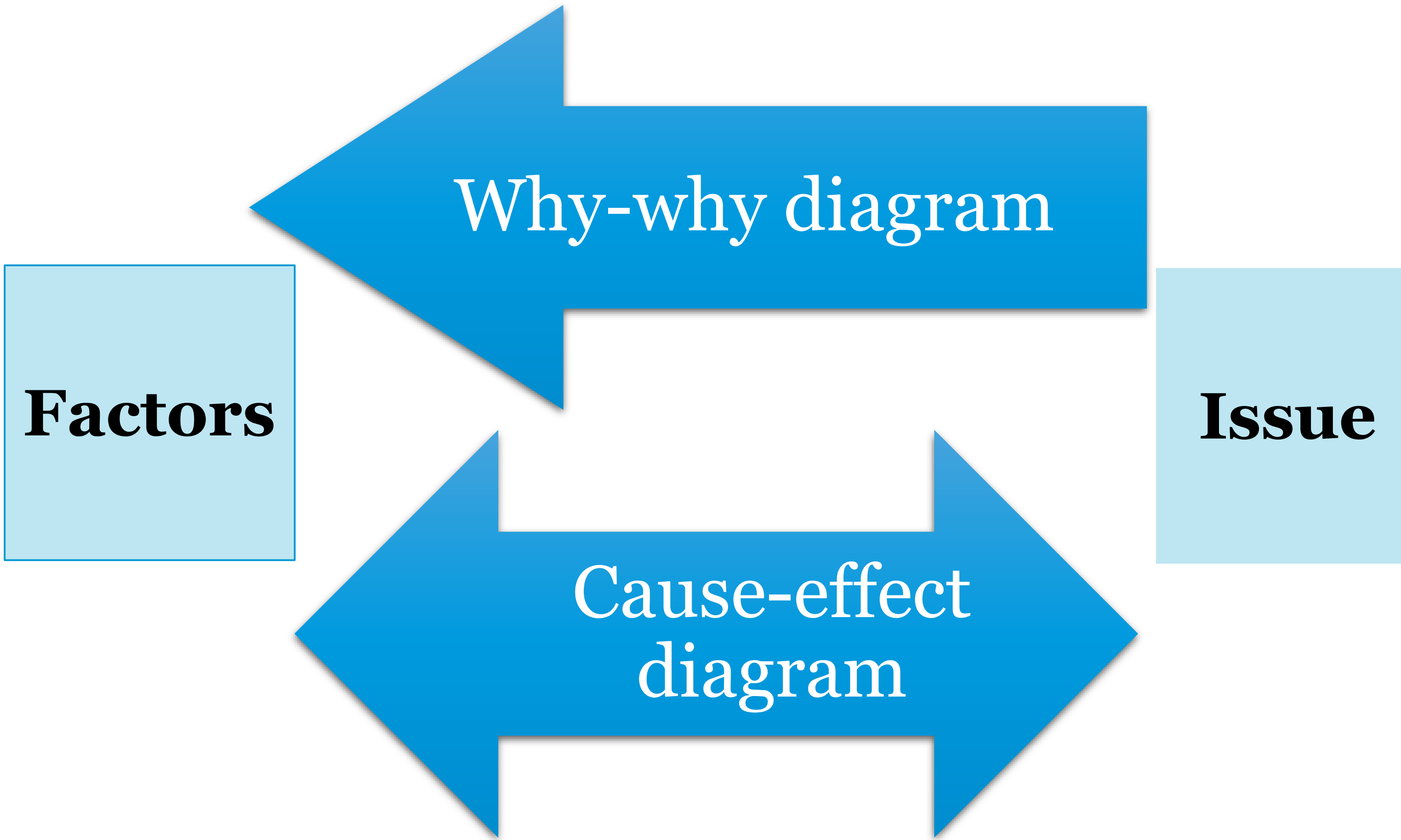
Dumas · La Rosa · Mendling · Reijers

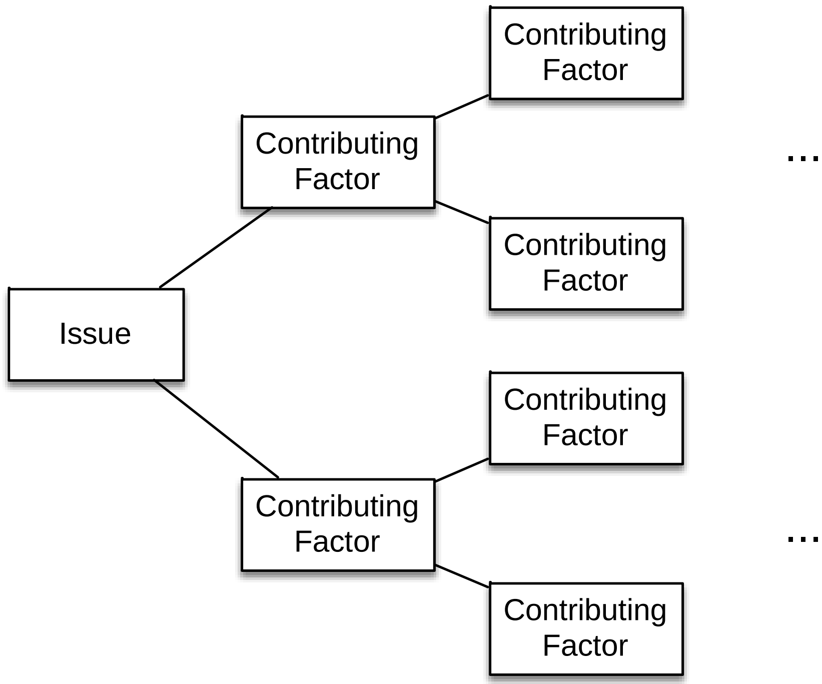
Fundamentals of Business Process Management

2nd Ed.

Root-cause analysis



Why-why diagram



Five levels of nesting

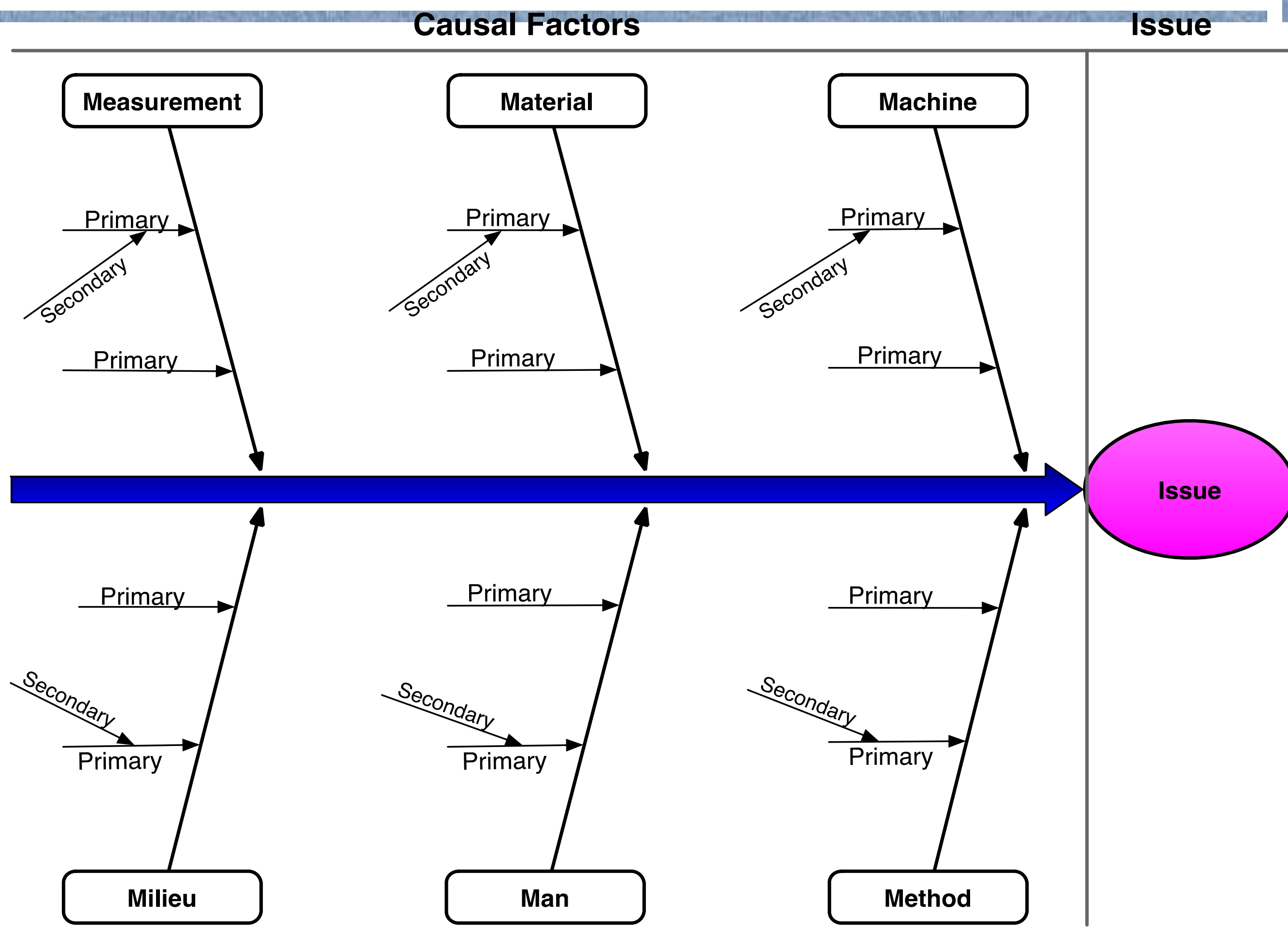
"Five Why's"

Why-why diagram example

Site engineers keep equipment longer, why?

- Site engineer fears that equipment will not be available later when needed, why?
  - time between request and delivery too long, why?
    - excessive time spent in finding suitable equipment and approving the request, why?
      - time spent by clerk contacting possibly multiple suppliers sequentially;
      - time spent waiting for works engineer to check the requests;

Cause-effect (Fishbone) diagram



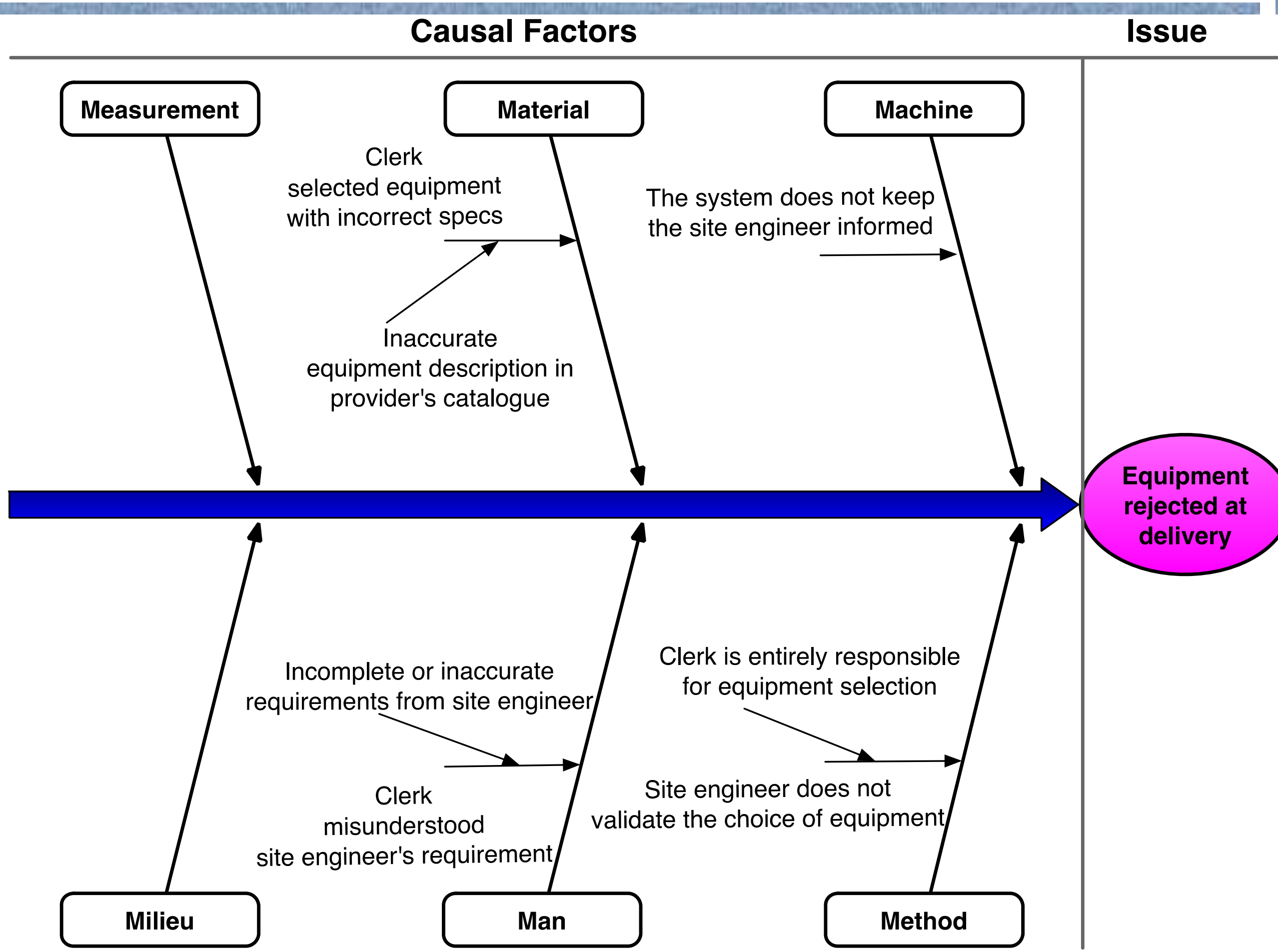
Categories of causes: Six Ms

1. **Machine:** factors stemming from technology used
   - Lack of suitable functionality in the supporting software applications
   - Poor User Interface (UI) design
   - Lack of integration between systems
2. **Method:** factors stemming from the way the process is designed, understood or performed
   - Unclear assignments of responsibilities
   - Unclear instructions
   - Insufficient training
   - Lack of timely communication
3. **Material:** factors stemming from input materials or data
   - Missing, incorrect or outdated data

Categories of causes: Six Ms

4. **Man:** factors stemming from wrong assessments or incorrect performance of steps attributable to:
   - Lack of training and clear instructions
   - Lack of motivation
   - Too high demands towards process workers
5. **Measurement:** factors stemming from reliance on:
   - Inaccurate estimations
   - Miscalculations
6. **Milieu:** factors outside the scope of the process
   - Delays caused because of unresponsive external actors
   - Sudden increases of workload due to special circumstances

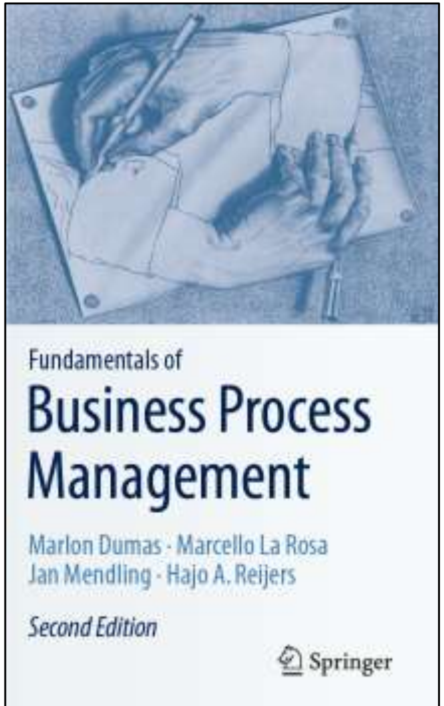
Cause-effect diagram example



Exercise

- Pharmacy prescription process (Exercise 1.6)
  - Select one of the previously identified issues and analyze it using a why-why diagram

Chapter 6: Qualitative Process Analysis



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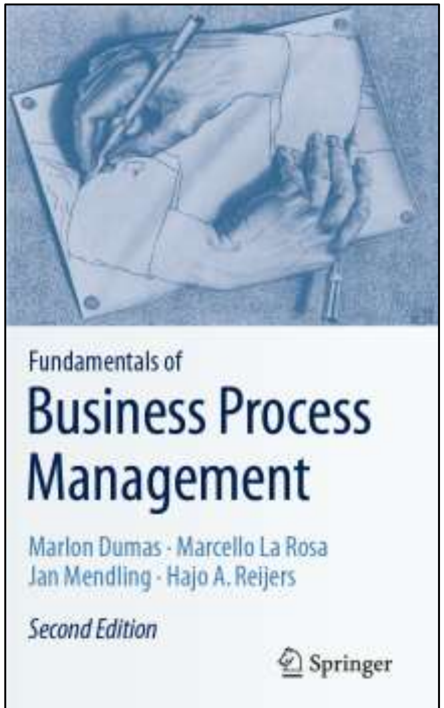
Recap

1. Segregate value-adding, business value-adding and non-value-adding steps
2. Identify waste: move, hold, overdo
3. Collect and systematically organize issues, assess their impact, priotirize
4. Analyze root causes of issues

Chapter 7: Quantitative Process Analysis



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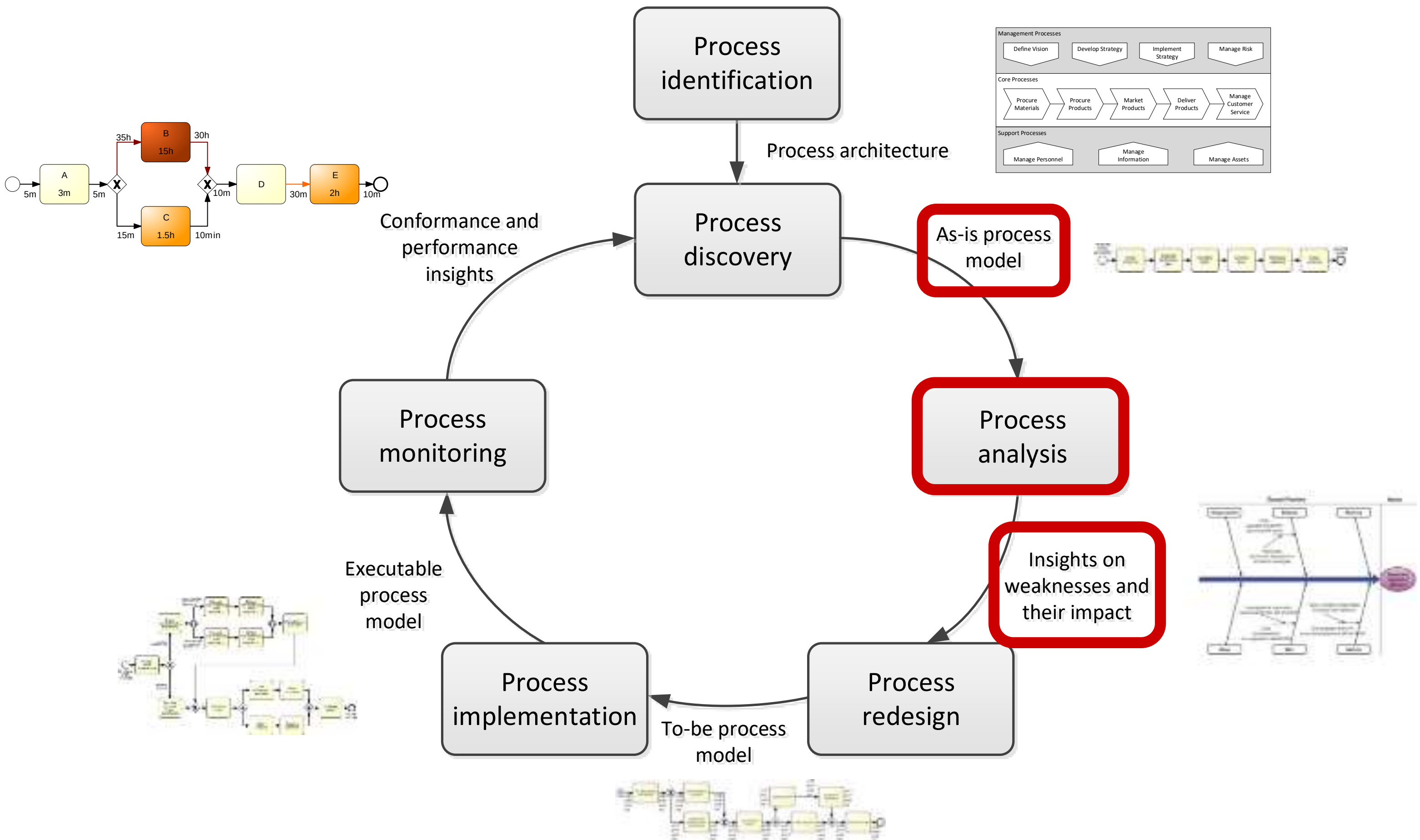
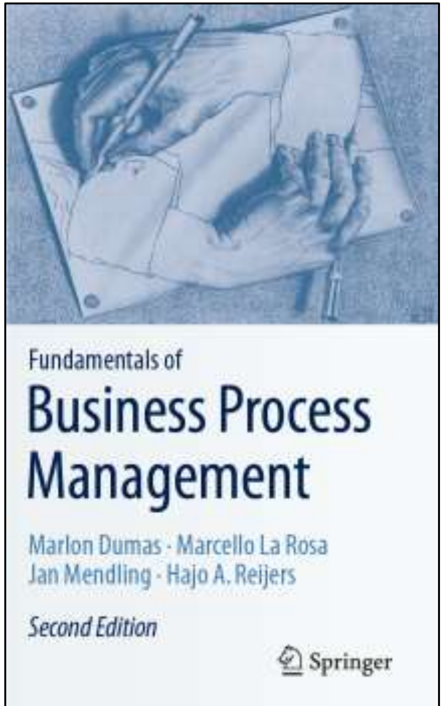
Process Analysis in the BPM Lifecycle



Chapter 7: Quantitative Process Analysis



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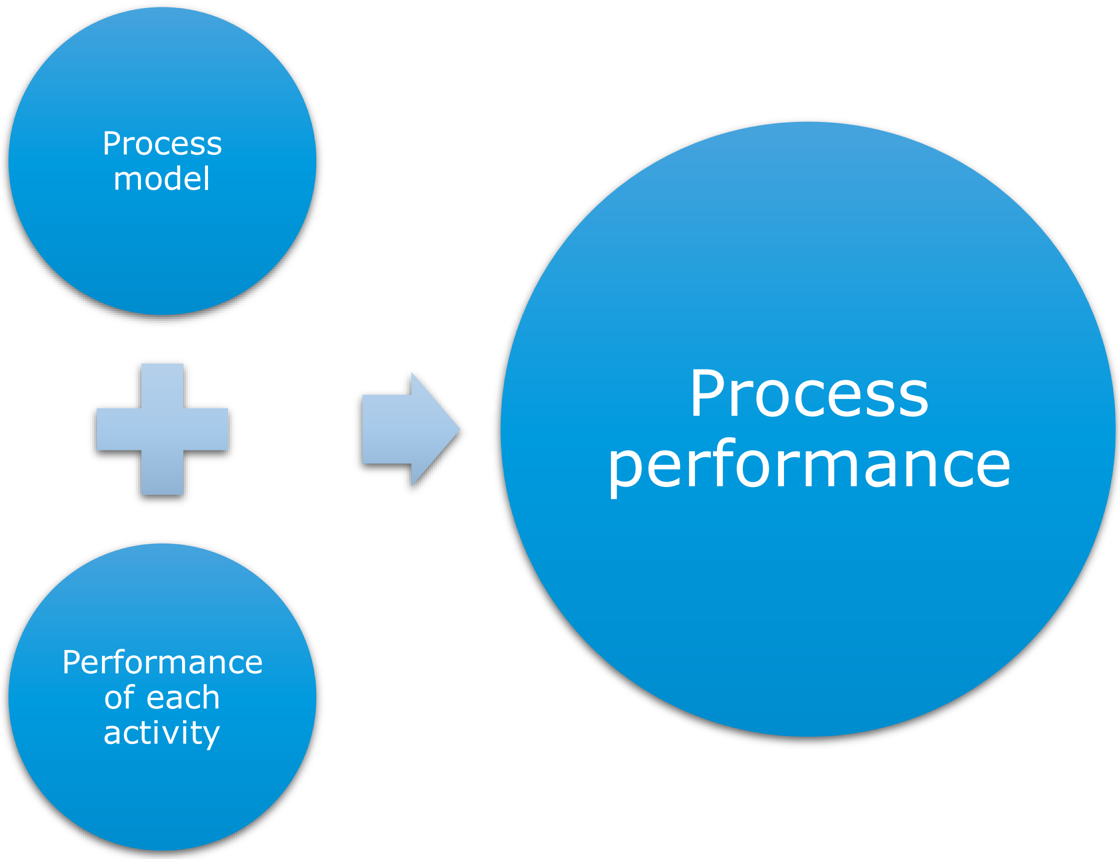
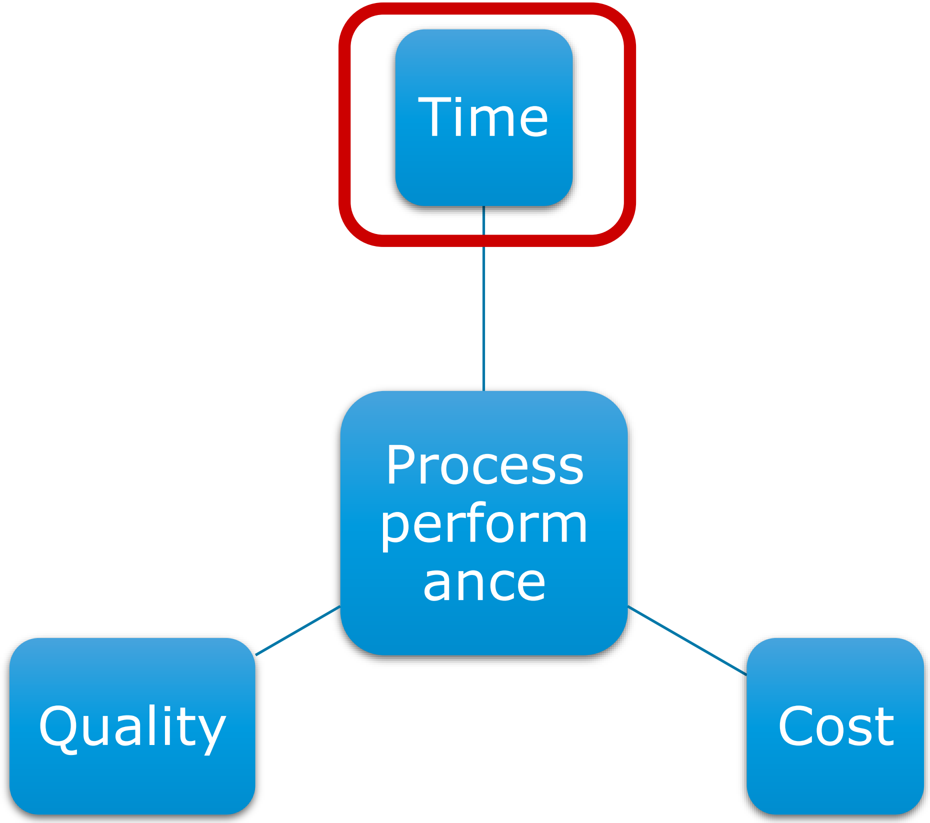
Flow analysis



Refresher: Process performance measures



Common time-related measures

Processing time

Time taken by value-adding activities

+

Waiting time

Time taken by non-value-adding activities

→

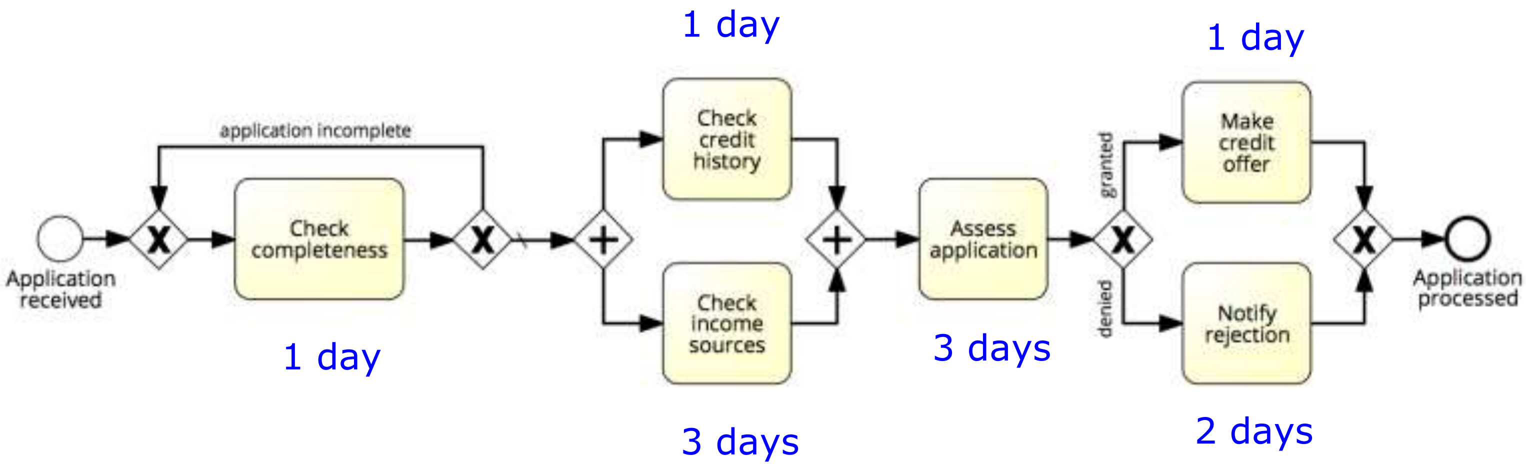
Cycle time

Time between start and completion of a process instance

Cycle time efficiency

Processing Time ÷ Cycle Time = Cycle Time Efficiency

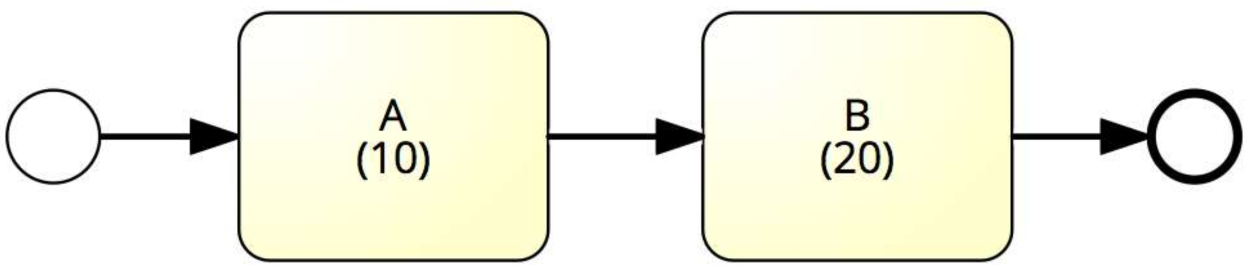
Flow analysis of cycle time



Cycle time = X days

Sequence – Example

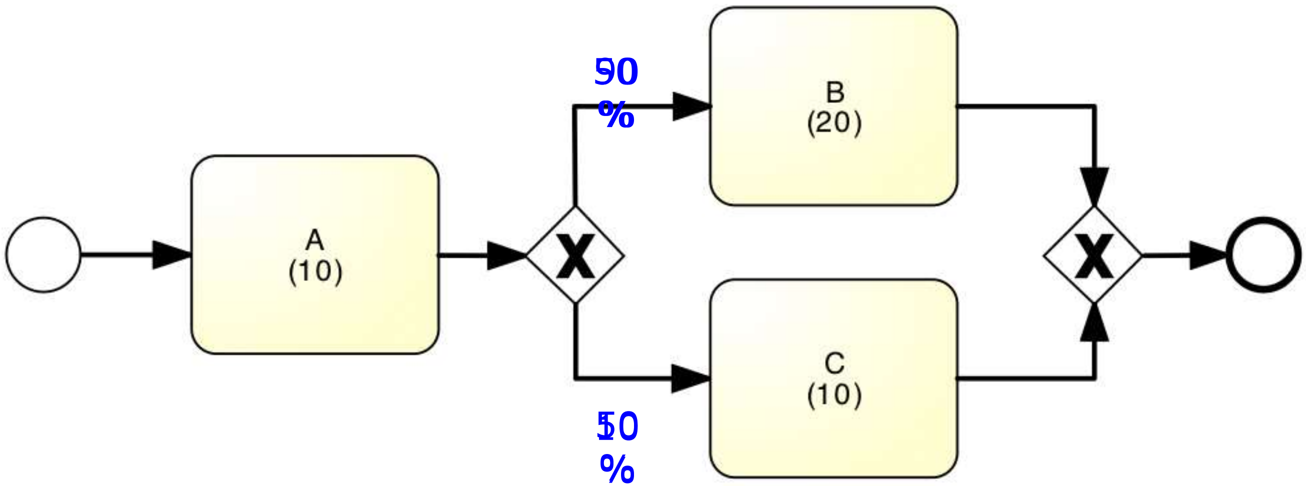
- What is the average cycle time?



Cycle time = 10 + 20 = 30

Example: Alternative Paths

- What is the average cycle time?

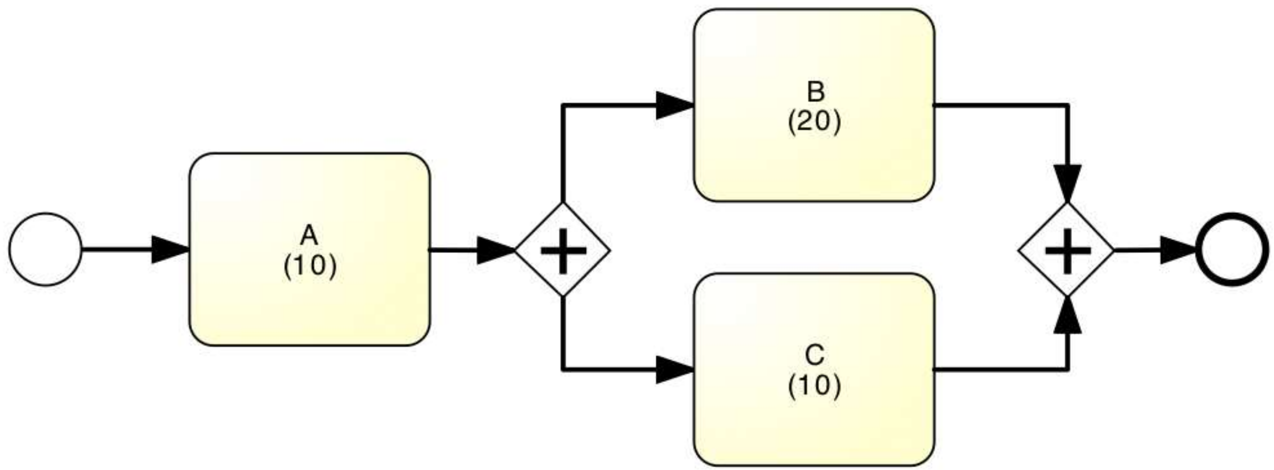


Cycle time = 10 + (20+10)/2 = 25

Cycle time = 10 + 0.9*20+0.1*10 = 29

Example: Parallel paths

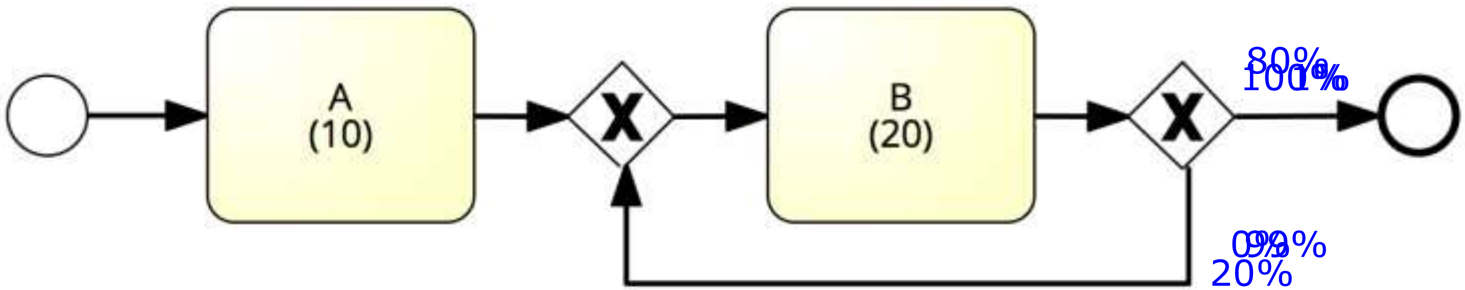
- What is the average cycle time?



Cycle time = 10 + 20 = 30

Example: Rework loop

- What is the average cycle time?

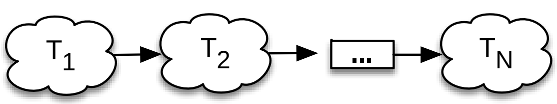


Cycle time = 10 + 20 = 30

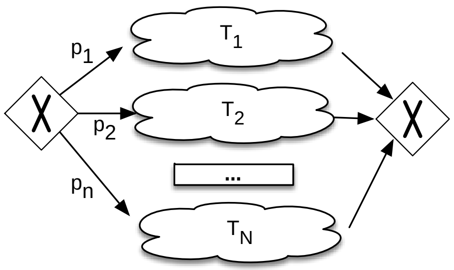
Cycle time = 10 + 20/0.01 = 2010

Cycle time = 10 + 20/0.8 = 35

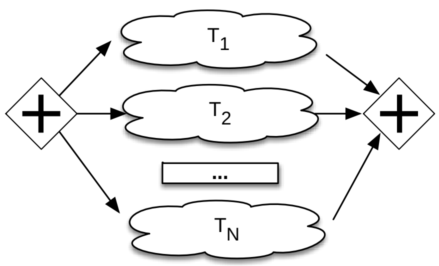
Flow analysis equations for cycle time



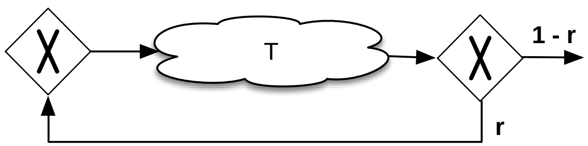
$$CT = T_1+T_2+\ldots+ T_N$$



$$CT = p_1*T_1+p_2*T_2+\ldots+ p_n*T_N$$

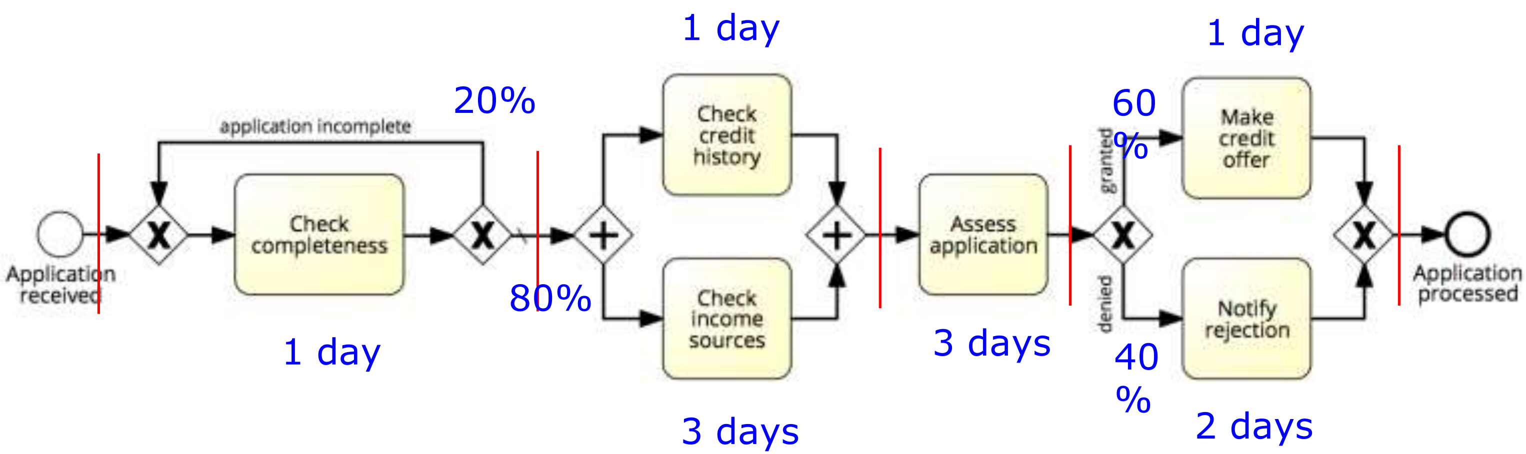


$$CT = \max(T_1, T_2,\ldots, T_N)$$



$$CT = T / (1-r)$$

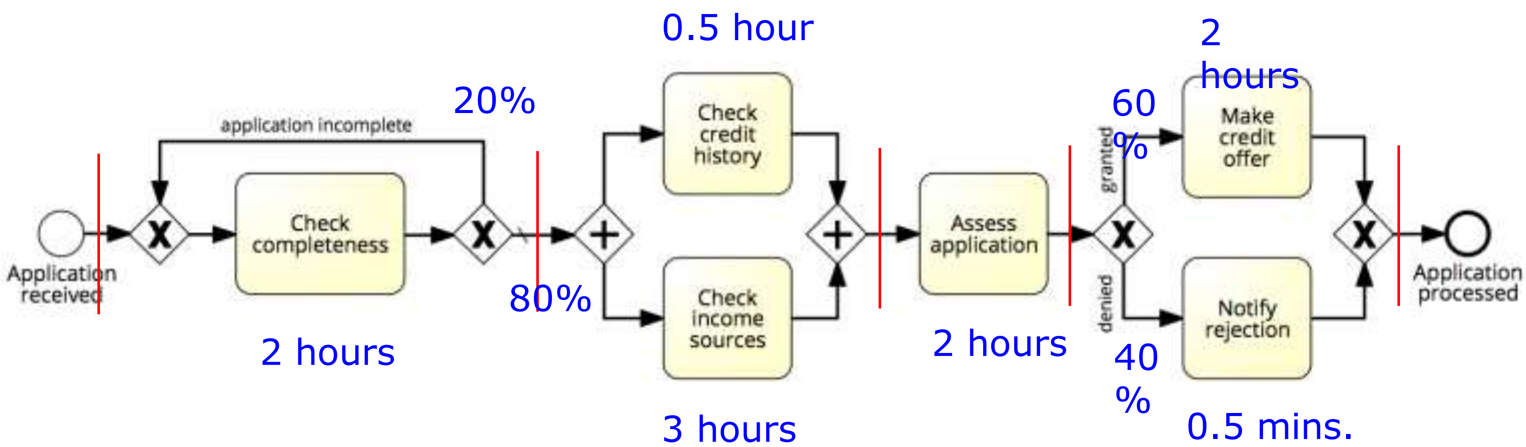
Flow analysis of cycle time



1/0.8 max(1,3) 3 0.6*1+0.4*2

Cycle time = 1.25 + 3 + 3 + 1.4 = 8.65 days

Flow analysis of processing time



2/0.8 max(0.5,3) 2 0.6*2+0.4*0.5

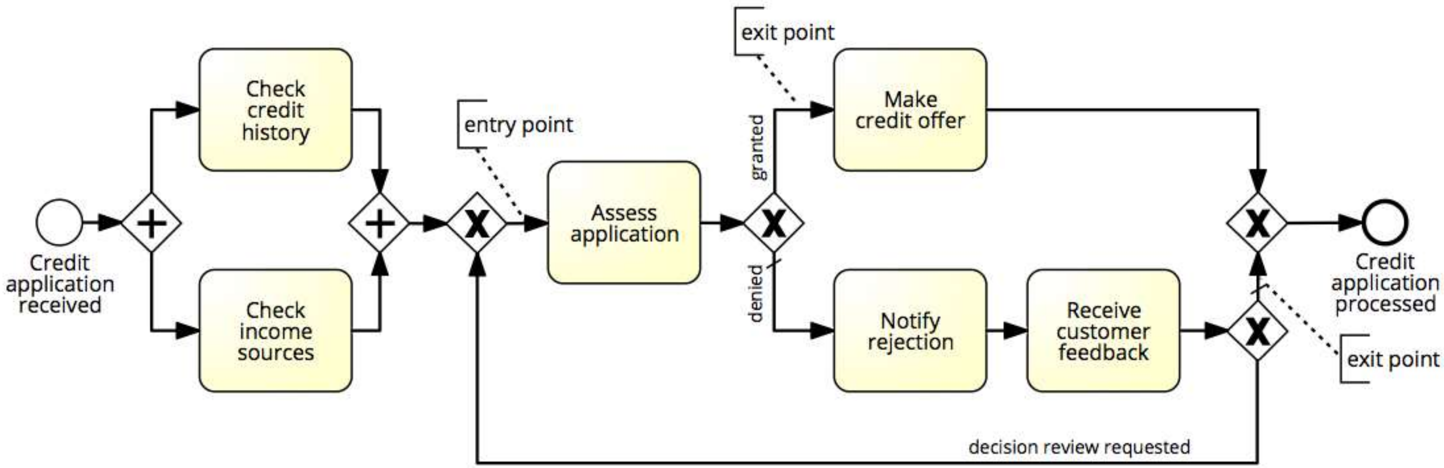
Processing time = 2.5 + 3 + 2 + 1.4 = 8.9 hours

Cycle time efficiency = 8.9 hours / 8.65 days = 12.9%

Flow analysis: scope and limitations

- We have seen how to use flow analysis for processing & cycle time calculation
- Flow analysis can also be applied to calculate:
  - The average cost of process instances (assuming we know the cost of each activity)
    - Cf. Section 7.1.6
  - The number of times on average each activity is executed
    - Can be used to calculate the “unit load” of each task, the resource utilization of each resource pool, and the theoretical capacity of an “as is” process
    - Cf. Section 7.1.5
- But flow analysis has some fundamental limitations…

Limitation 1: Not all Models are Structured



Limitation 2: Fixed arrival rate capacity

- Cycle time analysis does not consider:
  - The rate at which new process instances are created (arrival rate)
  - The number of available resources
- Higher arrival rate at fixed resource capacity
  ➔ high resource contention
  ➔ higher activity waiting times (longer queues)
  ➔ higher activity cycle time
  ➔ higher overall cycle time
- The slower you are, the more people have to queue up…
  - and vice-versa

Resource utilization

Time spent per resource on process work ÷ Time available per resource for process work = Resource utilization

Resource utilization = 60%
➔ on average resources are idle 40% of their allocated time

Resource utilization vs. waiting time

Typically, when resource utilization > 90%
➔ Waiting time increases steeply

Interlude: Cycle Time & Work-In-Progress

- WIP = (average) Work-In-Process
  - Number of cases that are running (started but not yet completed)
  - E.g. # of active and unfilled orders in an order-to-cash process
- WIP is a form of waste (cf. 7+1 sources of waste)
- Little's Formula: WIP = $\lambda \cdot$CT
  - λ = arrival rate (number of new cases per time unit)
  - CT = cycle time

Exercise

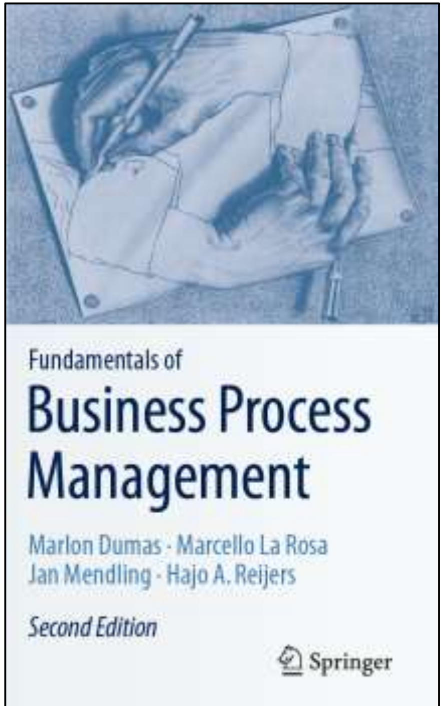
A fast-food restaurant receives on average 1200 customers per day (between 10:00 and 22:00). During peak times (12:00-15:00 and 18:00-21:00), the restaurant receives around 900 customers in total, and 90 customers can be found in the restaurant (on average) at a given point in time. At non-peak times, the restaurant receives 300 customers in total, and 30 customers can be found in the restaurant (on average) at a given point in time.

1. What is the average time that a customer spends in the restaurant during peak times?
2. What is the average time that a customer spends in the restaurant during non-peak times?

Exercise (cont.)

3. The restaurant plans to launch a marketing campaign to attract more customers. However, the restaurant's capacity is limited and becomes too full during peak times. What can the restaurant do to address this issue without investing in extending its building?

Chapter 7: Quantitative Process Analysis



Contents

1. Flow Analysis
2. Queuing Analysis
3. Simulation
4. Recap

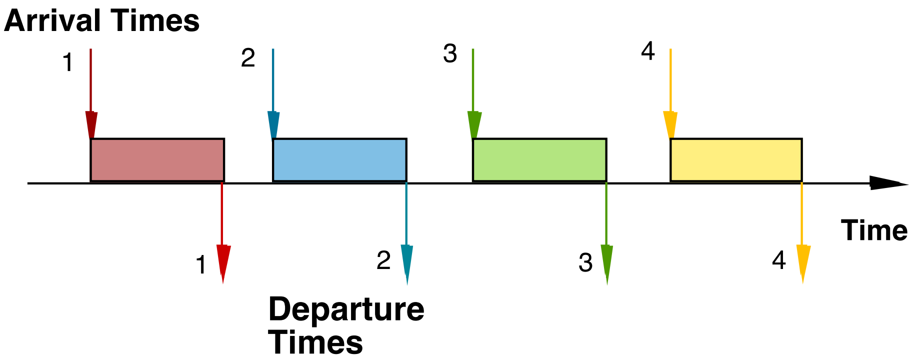
Queuing Analysis

- Capacity problems are common and a key driver of process redesign
  - Need to balance the cost of increased capacity against the gains of increased productivity and service
- Queuing and waiting time analysis is particularly important in service systems
  - Large costs of waiting and/or lost sales due to waiting
- **Example – Emergency Room (ER) at a Hospital**
  - Patients arrive by ambulance or by their own accord
  - One doctor is always on duty
  - More patients seeks help $\Rightarrow$ longer waiting times
  - *Should we increase the capacity from one to two doctors?*

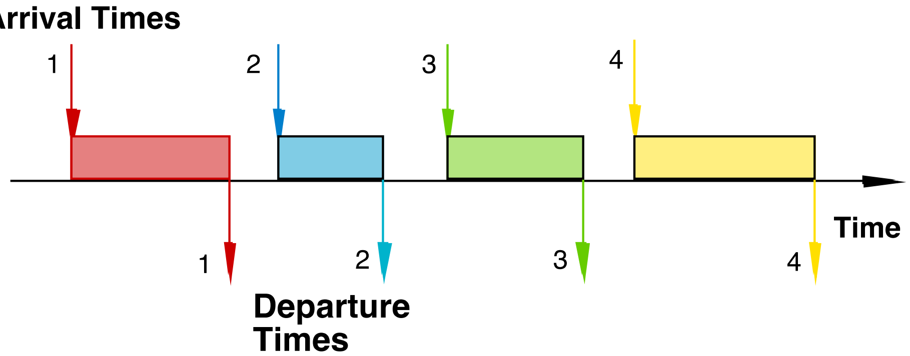
Inspired by an example by Laguna & Marklund (2004)

Delay is Caused by Job Interference

If arrivals are regular or sufficiently spaced apart, no queuing delay occurs



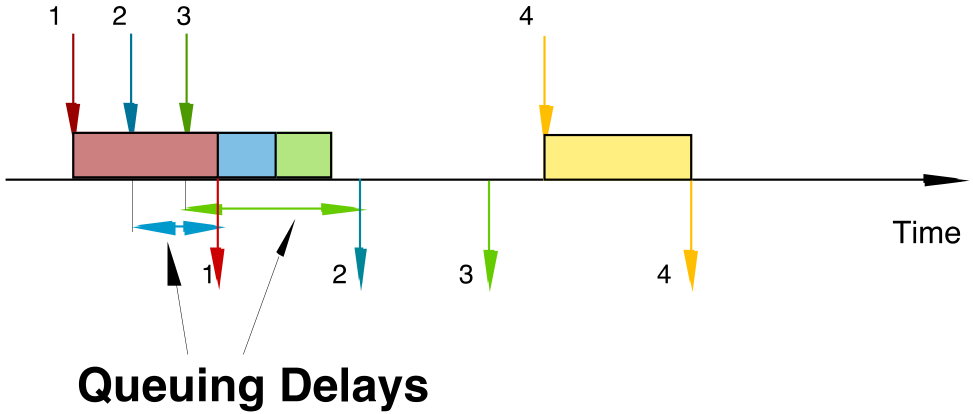
Deterministic traffic



Variable but spaced apart traffic

Burstiness Causes Interference

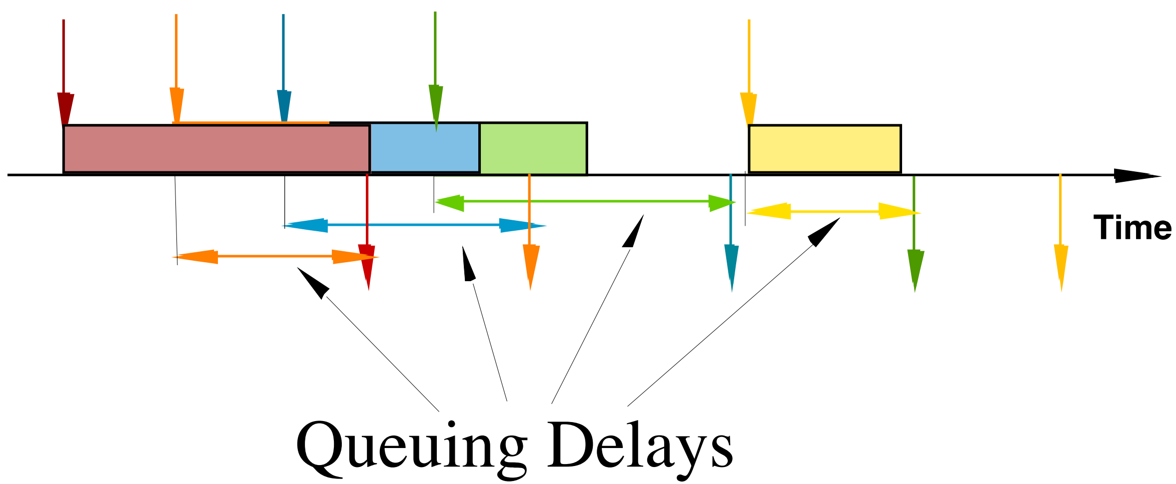
* Queuing results from variability in processing times and/or interarrival times



Bursty Traffic

High Utilization Exacerbates Interference

- The queuing probability increases as the load increases
- Utilization close to 100% is unsustainable → too long queuing times

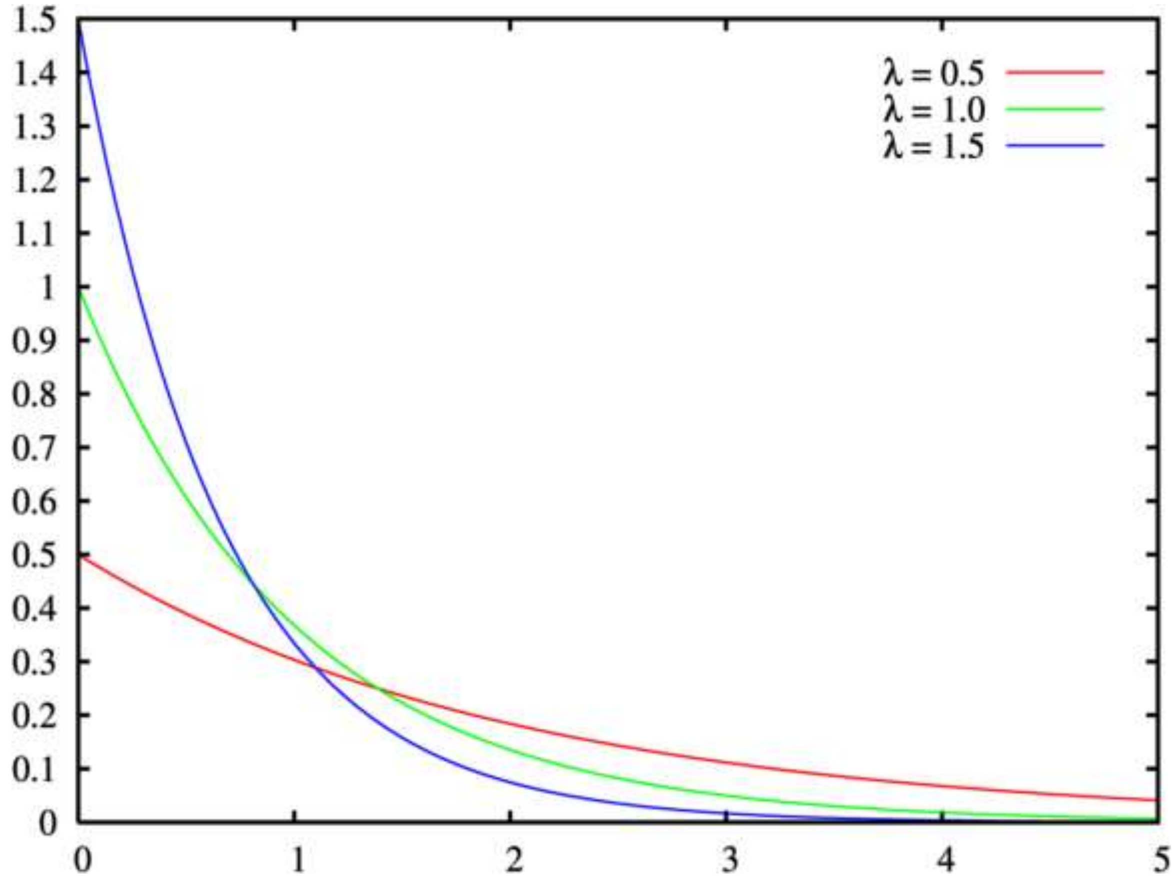


The Poisson Process

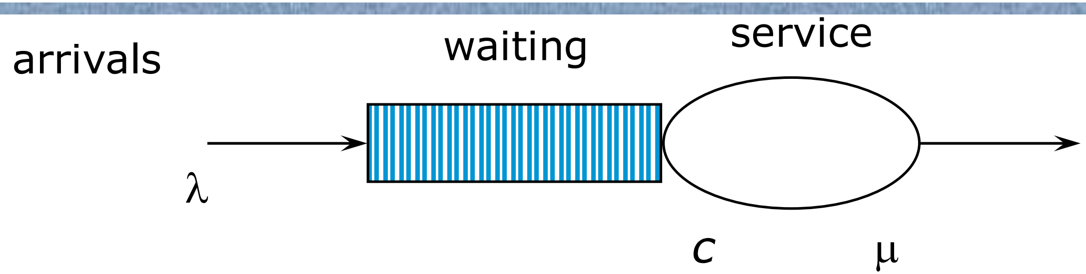
- Common arrival assumption in many queuing and simulation models
- The times between arrivals are independent, identically distributed and exponential
  - P (arrival < t) = $1 - e^{-\lambda t}$
- This distribution is applicable when the next arrival (i.e. the next creation of a case) does not depend on how long ago the previous arrival occurred
  - In other words, the creation of a case is independent of the creation of other cases.

Inspired by slide by Laguna & Marklund (2004)

Negative Exponential Distribution



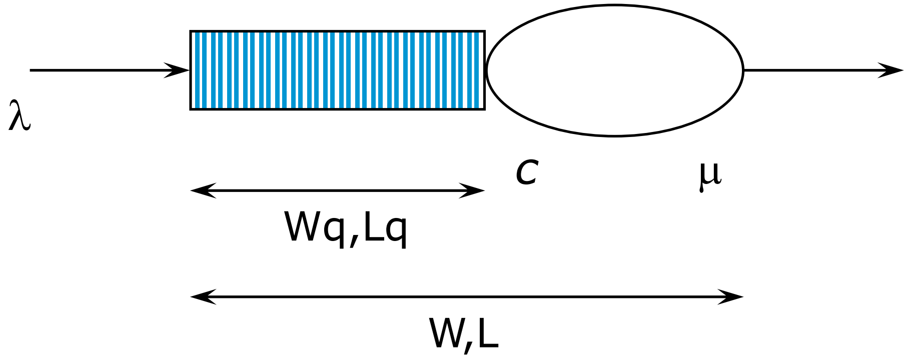
Queuing theory: basic concepts



Basic characteristics:

- λ (mean arrival rate) = average number of arrivals per time unit
- μ (mean service rate) = average number of jobs that can be handled by one server per time unit:
- c = number of servers

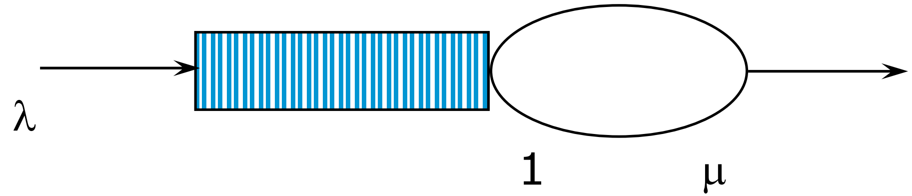
Queuing theory concepts (cont.)



Given λ , μ and c, we can calculate :

- **occupation rate: ρ**
- **Wq = average time in queue**
- **W = average system in system (i.e. cycle time)**
- **Lq = average number in queue (i.e. length of queue)**
- **L = average number in system average (i.e. Work-in-Progress)**

M/M/1 queue



Assumptions:

- time between arrivals and processing time follow a negative exponential distribution
- 1 server (c = 1)
- FIFO

$$\rho = \frac{\text{Capacity Demand}}{\text{Available Capacity}} = \frac{\lambda}{\mu}$$

$L = \rho/(1-\rho)$

$W = L/\lambda = 1/(\mu-\lambda)$

$L_q = \rho^2/(1-\rho) = L-\rho$

$W_q = L_q/\lambda = \lambda/(\mu(\mu-\lambda))$

Inspired by a slide by Laguna & Marklund (2004)

M/M/c queue

- Now there are c servers in parallel, so the expected capacity per time unit is then $c*\mu$

$$\rho = \frac{\text{Capacity Demand}}{\text{Available Capacity}} = \frac{\lambda}{c*\mu}$$

Little's Formula $\Rightarrow$ $\mathbf{W_q=L_q/\lambda}$

$\mathbf{W=W_q+(1/\mu)}$

Little's Formula $\Rightarrow$ $\mathbf{L=\lambda W}$

Inspired by a slide by Laguna & Marklund (2004)

Tool Support

- For M/M/c systems, the exact computation of L_q is rather complex…

$$L_q = \sum_{n=c}^{\infty}(n-c)P_n = \ldots = \frac{(\lambda/\mu)^c \rho}{c!(1-\rho)^2} P_0$$

$$P_0 = \left(\sum_{n=0}^{c-1} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^c}{c!} \cdot \frac{1}{1-(\lambda/(c\mu))} \right)^{-1}$$

- Consider using a tool, e.g.
  - http://www.supositorio.com/rcalc/rcalclite.htm
  - http://queueingtoolpak.org/ (for Excel)

Example – ER at County Hospital

- Situation
  - Patients arrive according to a Poisson process with intensity λ ($\Leftrightarrow$ the time between arrivals is exp(λ) distributed.
  - The service time (the doctor's examination and treatment time of a patient) follows an exponential distribution with mean $1/\mu$ (=exp(μ) distributed)
  - $\Rightarrow$ *The ER can be modeled as an M/M/c system where c = the number of doctors*
- Data gathering
  - $\Rightarrow$ λ = 2 patients per hour
  - $\Rightarrow$ μ = 3 patients per hour
- Question
- – Should the capacity be increased from 1 to 2 doctors?

Inspired by a slide by Laguna & Marklund (2004)

Queuing Analysis – Hospital Scenario

- Interpretation
  - To be in the queue = to be in the waiting room
  - To be in the system = to be in the ER (waiting or under treatment)

Characteristic	One doctor (c=1)	Two Doctors (c=2)
ρ	2/3	1/3
L_q	4/3 patients	1/12 patients
L	2 patients	3/4 patients
W_q	2/3 h = 40 minutes	1/24 h = 2.5 minutes
W	1 h	3/8 h = 22.5 minutes

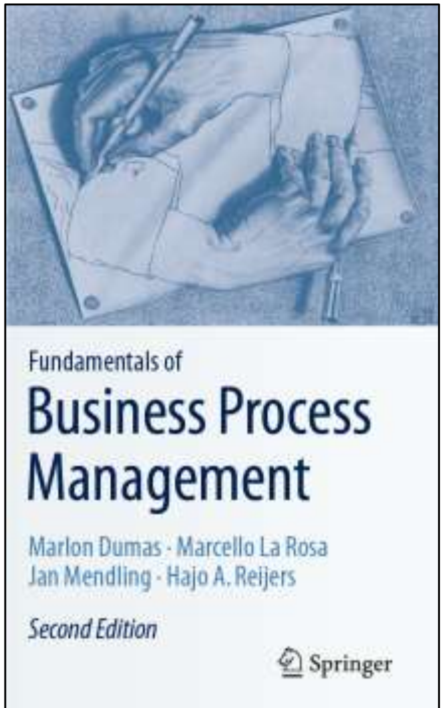
- Should we increase the capacity from one to two doctors?

Inspired by a slide by Laguna & Marklund (2004)

Limitations of basic queuing models

- Can be used to analyze waiting times (and hence cycle times), but not cost or quality measures
- Suitable for analyzing one single activity at a time, performed by one single resource pool. Not suitable for analyzing end-to-end processes consisting of multiple activities performed by multiple resource pools.
- These limitations are addressed by process simulation

Chapter 7: Quantitative Process Analysis



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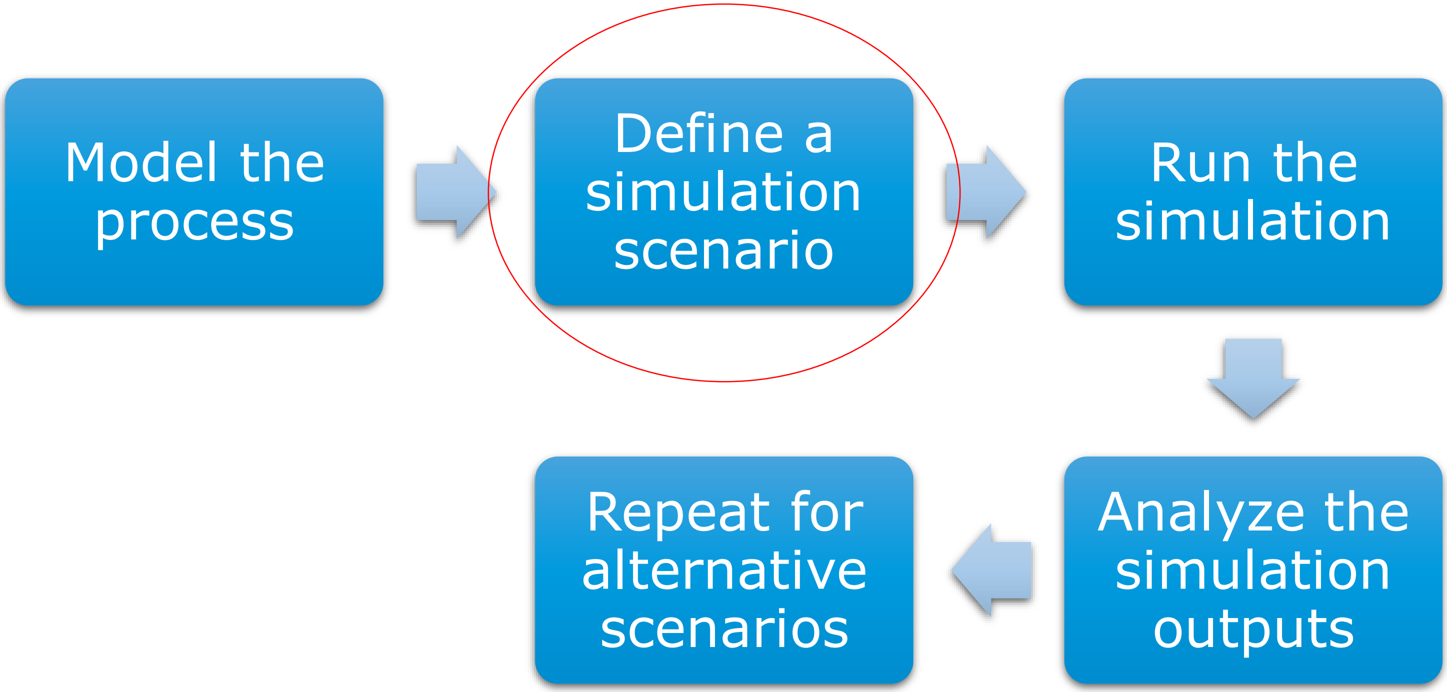
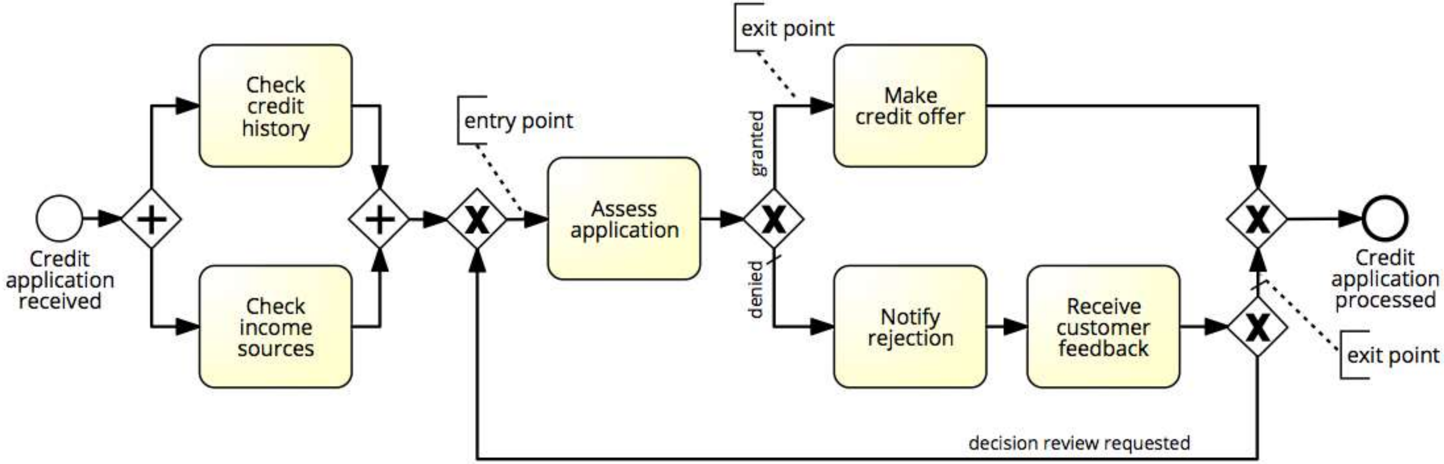
Process Simulation

- Versatile quantitative analysis method for
  - As-is analysis
  - What-if analysis
- In a nutshell:
  - Run a large number of process instances
  - Gather performance data (cost, time, resource usage)
  - Calculate statistics from the collected data

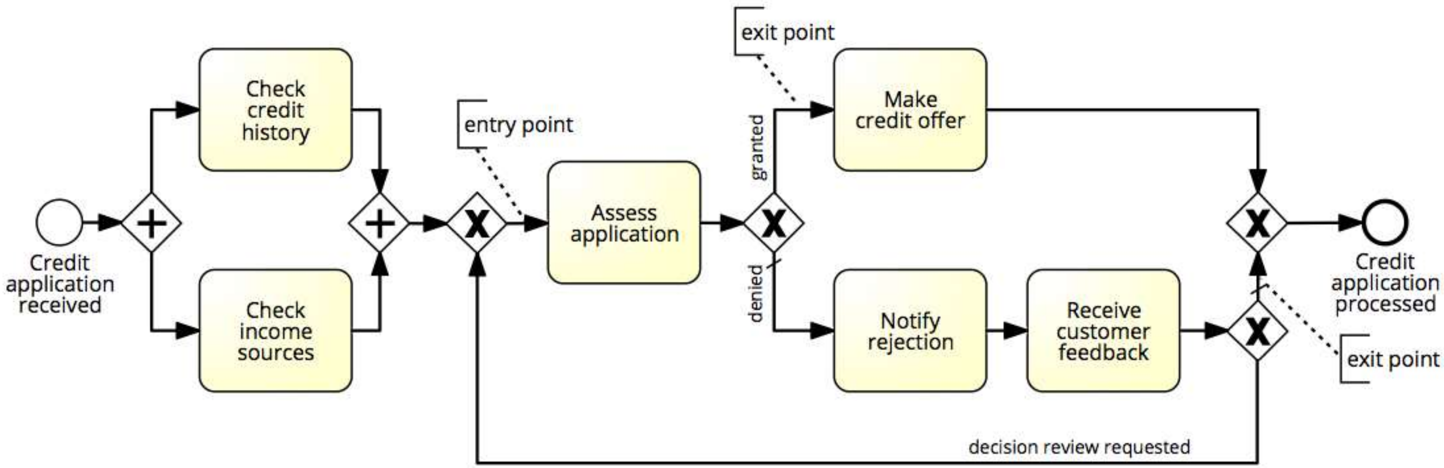
Process Simulation



Example


exit point
entry point
Check credit history
Check income sources
Credit application received
Assess application
granted
denied
Make credit offer
Notify rejection
Receive customer feedback
Credit application processed
exit point
decision review requested

Example

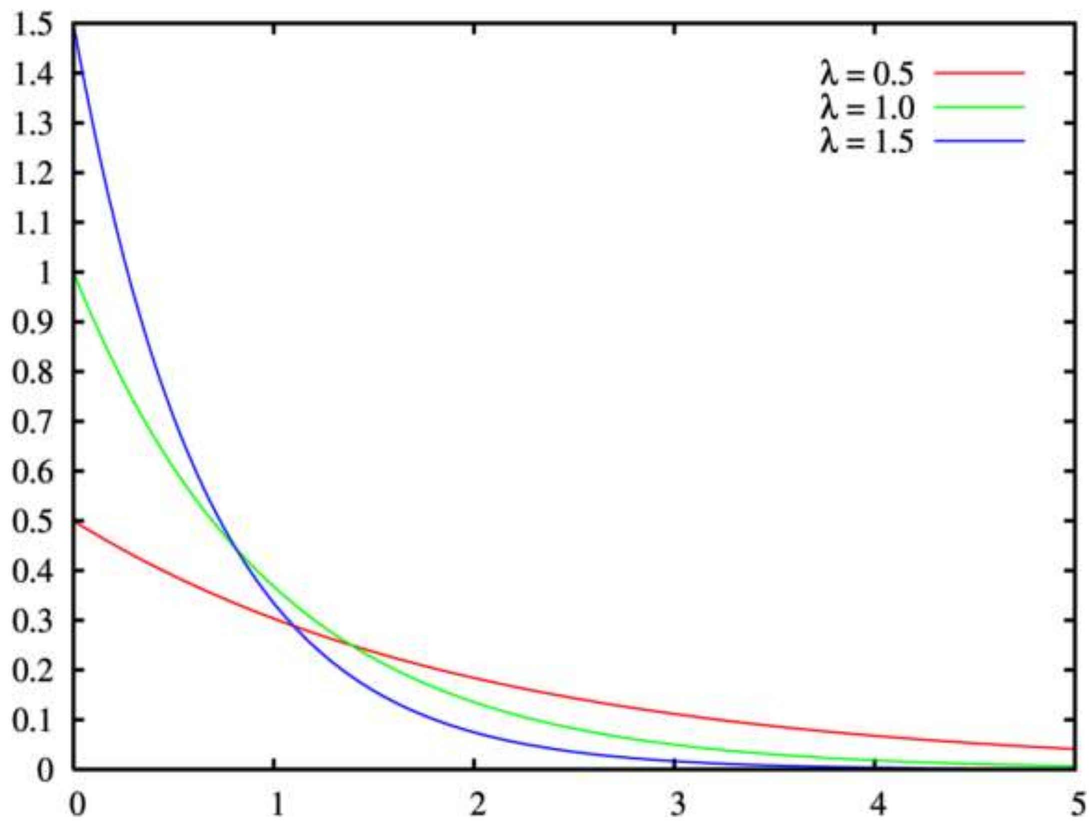


Credit application received
Check credit history
Check income sources
entry point
Assess application
exit point
granted
denied
Make credit offer
Notify rejection
Receive customer feedback
exit point
Credit application processed
decision review requested

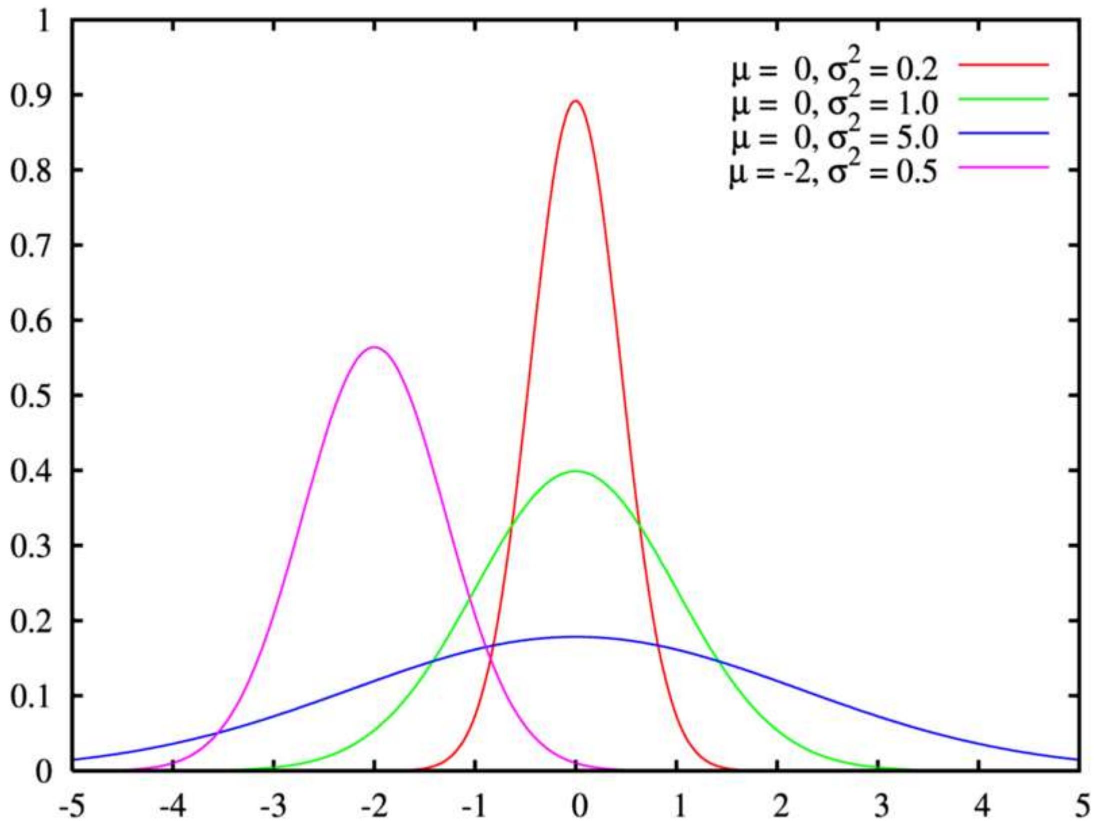
Elements of a simulation scenario

1. Processing times of activities
   - Fixed value
   - Probability distribution

Exponential Distribution



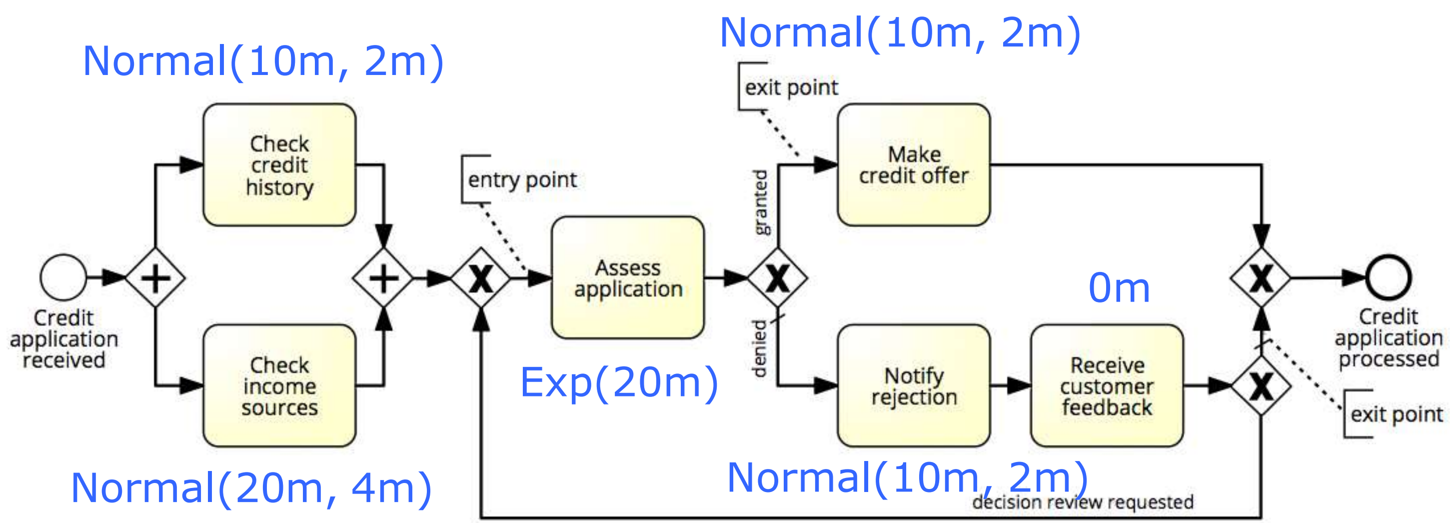
Normal Distribution



Choice of probability distribution

- Fixed
  - Rare, can be used to approximate case where the activity processing time varies very little
  - Example: a task performed by a software application
- Normal
  - Repetitive activities
  - Example: “Check completeness of an application”
- Exponential
  - Complex activities that may involve analysis or decisions
  - Example: “Assess an application”

Simulation Example

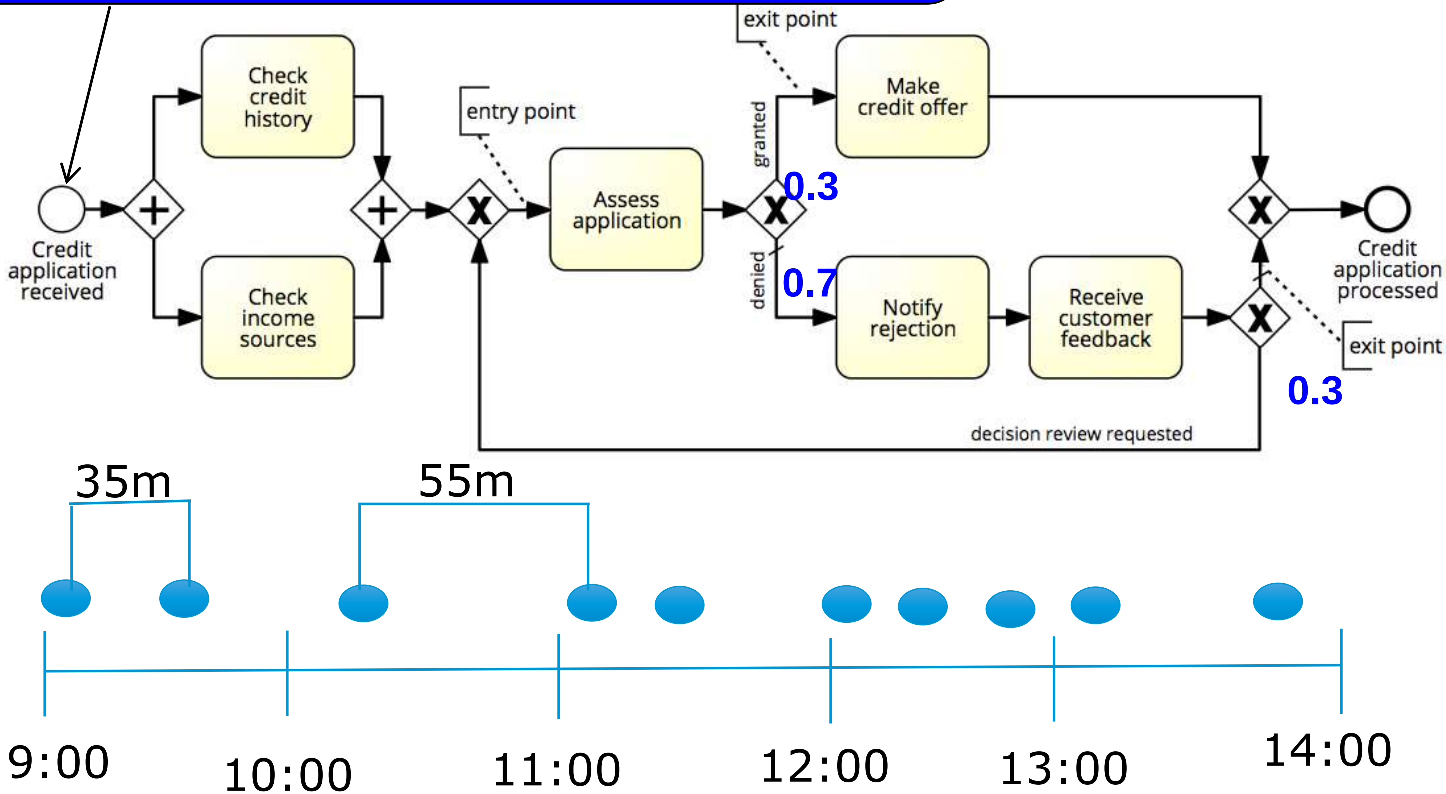


Elements of a simulation model

1. Processing times of activities
   - Fixed value
   - Probability distribution
2. Conditional branching probabilities
3. Arrival rate of process instances and probability distribution
   - Typically exponential distribution with a given mean inter-arrival time
   - Arrival calendar, e.g. Monday-Friday, 9am-5pm, or 24/7

Branching probability and arrival rate

Arrival rate = 2 applications per hour
Inter-arrival time = 0.5 hour
Negative exponential distribution
From Monday-Friday, 9am-5pm



Elements of a simulation model

1. Processing times of activities
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3. Arrival rate of process instances
   - Typically exponential distribution with a given mean inter-arrival time
   - Arrival calendar, e.g. Monday-Friday, 9am-5pm, or 24/7
4. Resource pools

Resource pools

- Name
- Size of the resource pool
- Cost per time unit of a resource in the pool
- Availability of the pool (working calendar)
- Examples

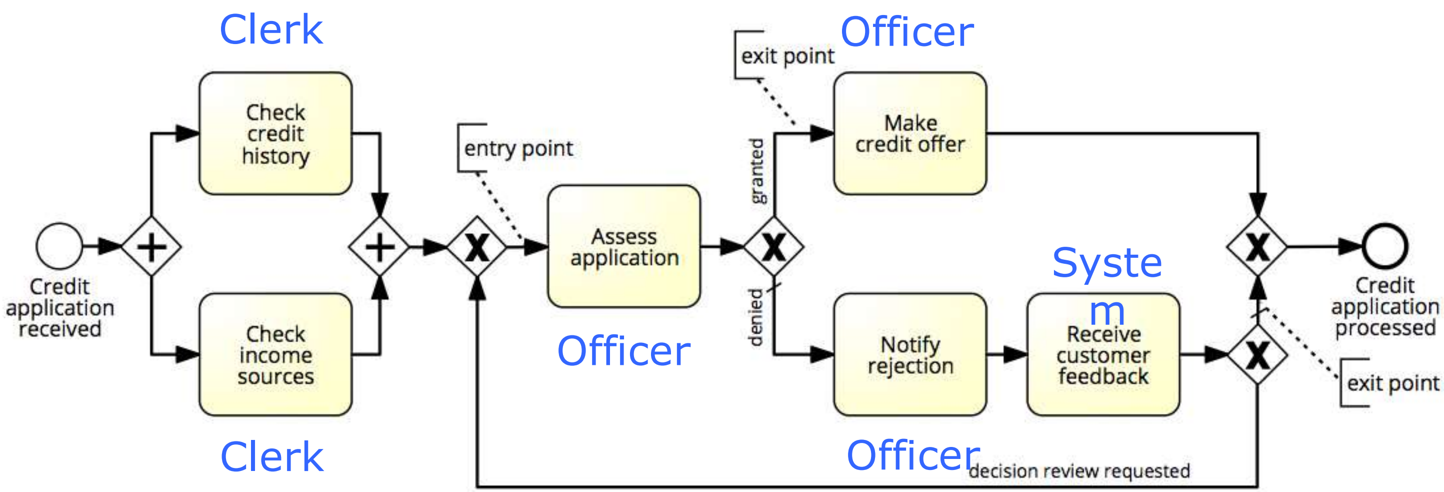
Clerk	Credit Officer
€ 25 per hour	€ 25 per hour
Monday-Friday, 9am-5pm	Monday-Friday, 9am-5pm

- In some tools, it is possible to define cost and calendar per resource, rather than for entire resource pool

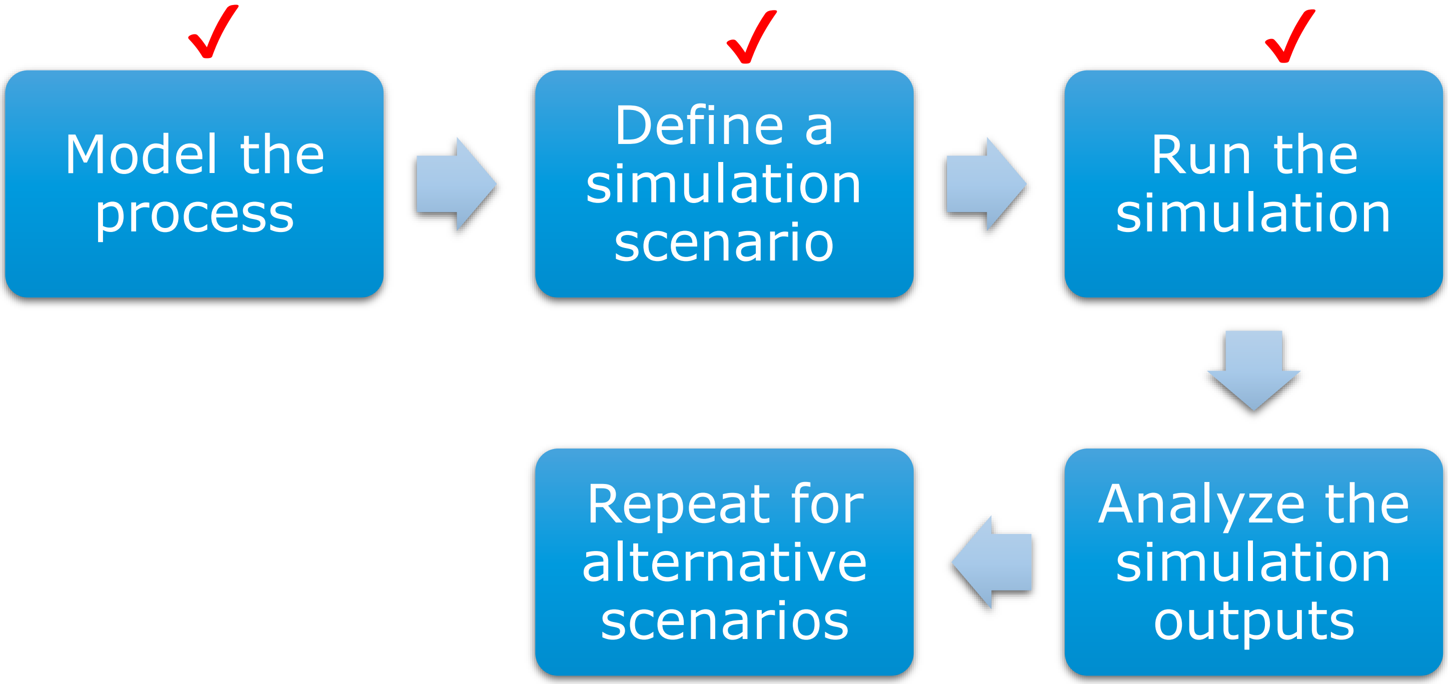
Elements of a simulation model

1. Processing times of activities
   - Fixed value
   - Probability distribution
2. Conditional branching probabilities
3. Arrival rate of process instances and probability distribution
   - Typically exponential distribution with a given mean inter-arrival time
   - Arrival calendar, e.g. Monday-Friday, 9am-5pm, or 24/7
4. Resource pools
5. Assignment of tasks to resource pools

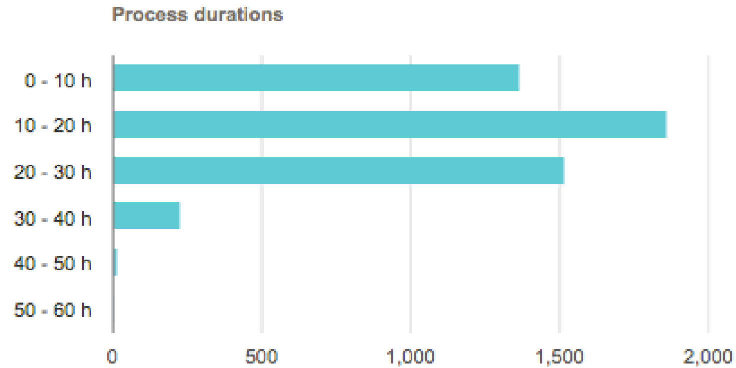
Resource pool assignment

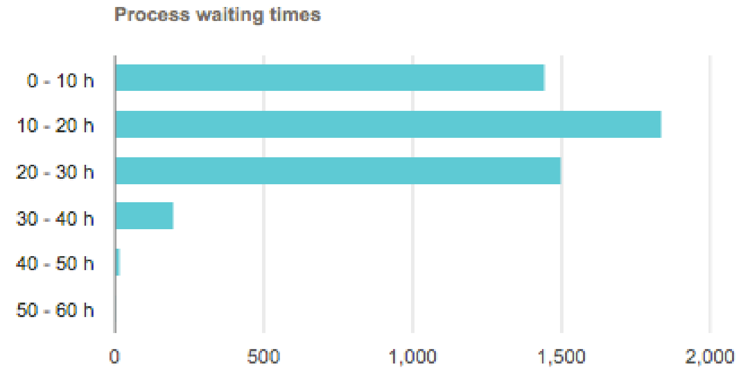


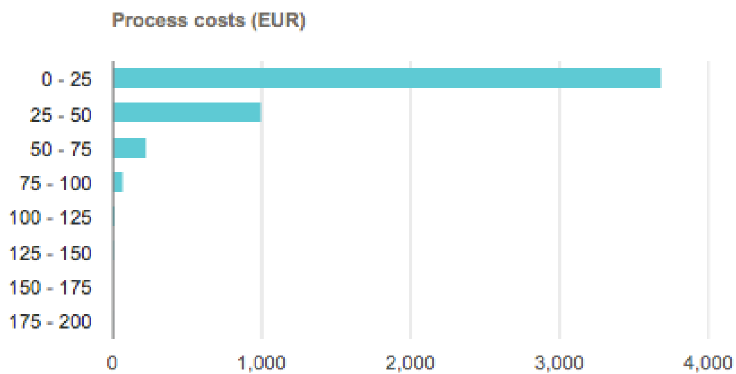
Process Simulation

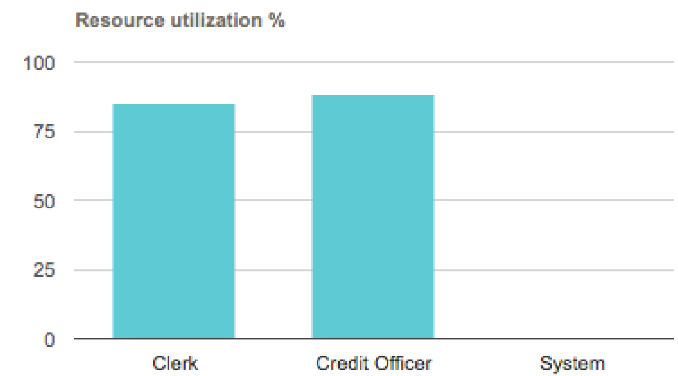


Output: Performance measures & histograms









Process Simulation

1. Model the process ✓
2. Define a simulation scenario ✓
3. Run the simulation ✓
4. Analyze the simulation outputs ✓
5. Repeat for alternative scenarios

Tools for Process Simulation

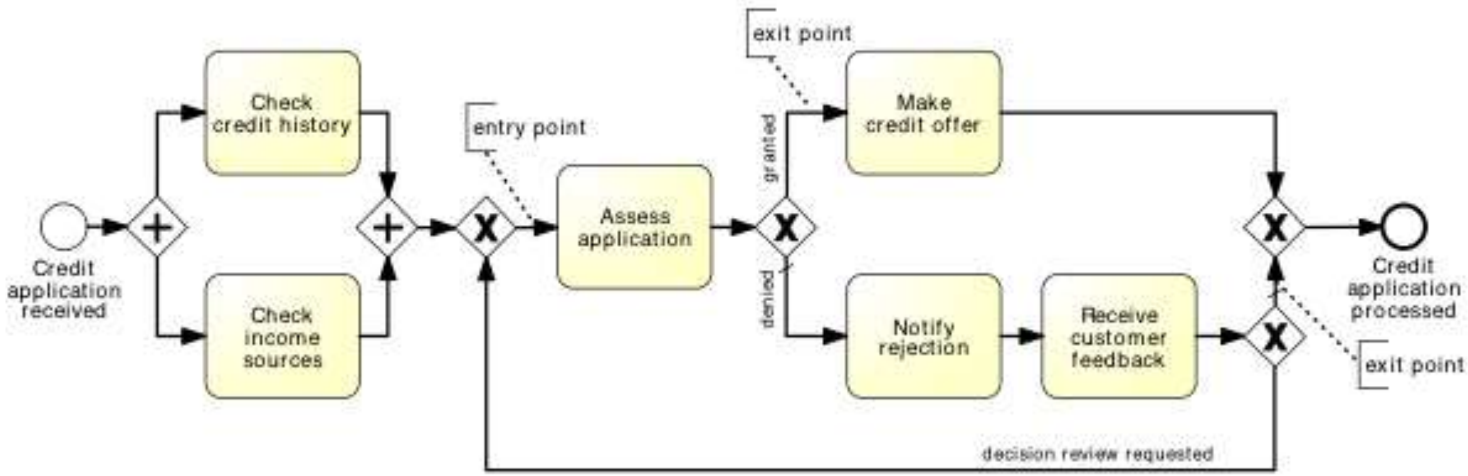
- ARIS
- Bizagi Process Modeler
- ITP Commerce Process Modeler for Visio
- Logizian
- Oracle BPA
- Progress Savvion Process Modeler
- ProSim
- Signavio + BIMP

BIMP – bimp.cs.ut.ee

- Accepts standard BPMN 2.0 as input
- Simple form-based interface to enter simulation scenario
- Produces KPIs + simulation logs in MXML format
  - Simulation logs can be imported to the ProM process mining tool

BIMP Demo

https://www.youtube.com/watch?v=TjXl6yASCSc



Pitfalls of simulation

- Stochasticity
- Data quality pitfalls
- Simplifying assumptions

Stochasticity

- Simulation results may differ from one run to another
- Make the simulation tiemframe long enough to cover weekly and seasonal variability, where applicable
- Use multiple simulation runs
- Average results of multiple runs, compute confidence intervals

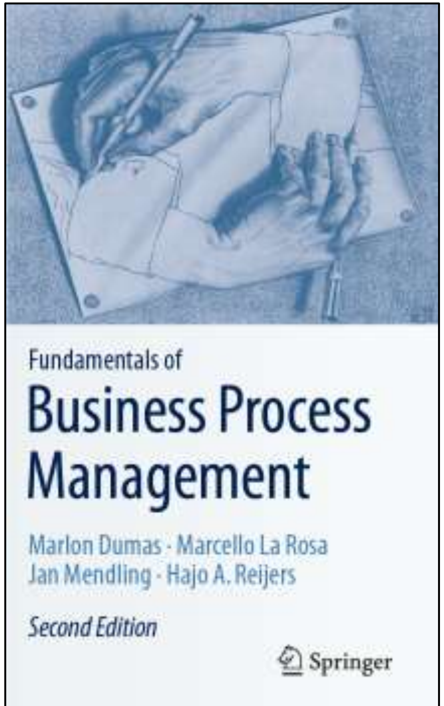
Data quality pitfalls

- Simulation results are only as trustworthy as the input data
- Rely as little as possible on “guesstimates”
- Use input analysis
  - Deriver simulation scenario parameters from numbers in the scenario
  - Use statistical tools to check fit the probability distributions
- Simulate the “as is” scenario and cross-check results against actual observations

Simulation assumptions

- That the process model is always followed to the letter
  - No deviations
  - No workarounds
- That there is no multi-tasking (the same resource performs multiple tasks concurrently) nor batching (tasks being accumulated and performed in a single go)
- That resources work constantly and non-stop
  - Every day is the same!
  - No tiredness effects
  - No distractions beyond “stochastic” ones

Chapter 7: Quantitative Process Analysis



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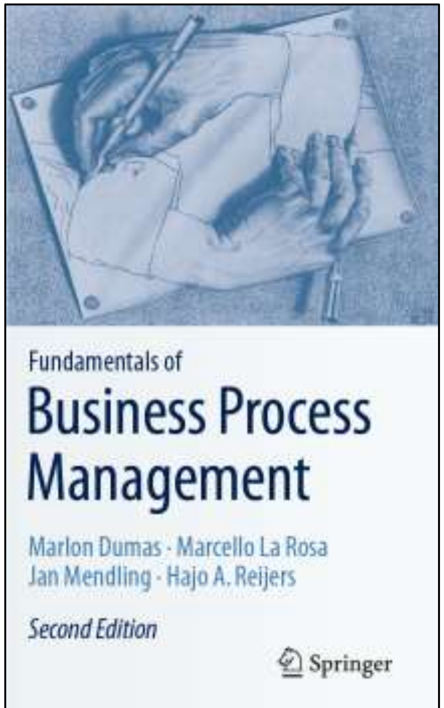
Recap

- Assuming we have performance measures for each activity in a process, flow analysis allows us to calculate the following performance measures for an <u>“as is” process:</u>
  - Cycle time, processing times, cycle time efficiency of a process
  - Average cost per process instance
- It can also be used to calculate the theoretical capacity of an “as is” process and the resource utilization of resource pools
- But it is not suitable for “what if” analysis
- Queing analysis is a suitable technique for “what if” analysis of waiting times and cycle times, suitable for analyzing individual activities performed by one resource pool
- Simulation is a versatile technique for “what if” analysis of entire processes, covering waiting times, cycle times, and costs.
  - Particularly useful for identifying bottlenecks

Chapter 8: Process Redesign



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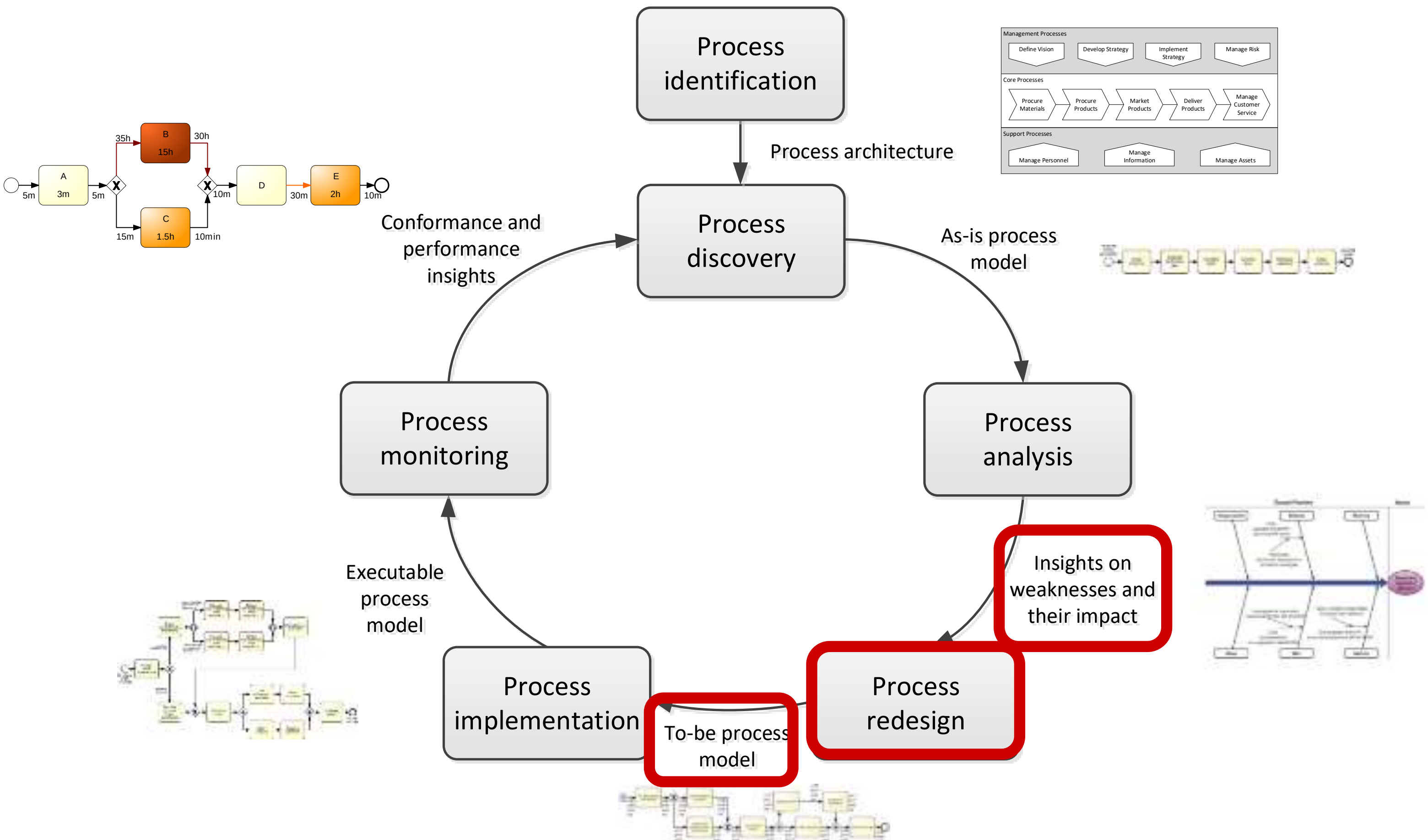
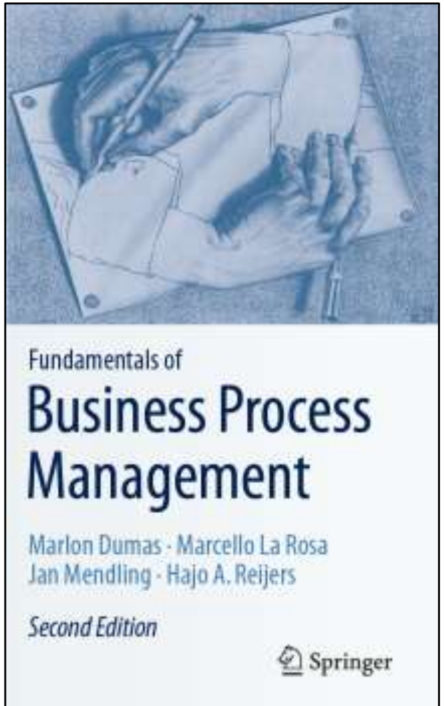
Process Redesign in the BPM Lifecycle



Chapter 7: Quantitative Process Analysis



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Dumas · La Rosa · Mendling · Reijers

Fundamentals of Business Process Management

2nd Ed.

Motives for Process Redesign

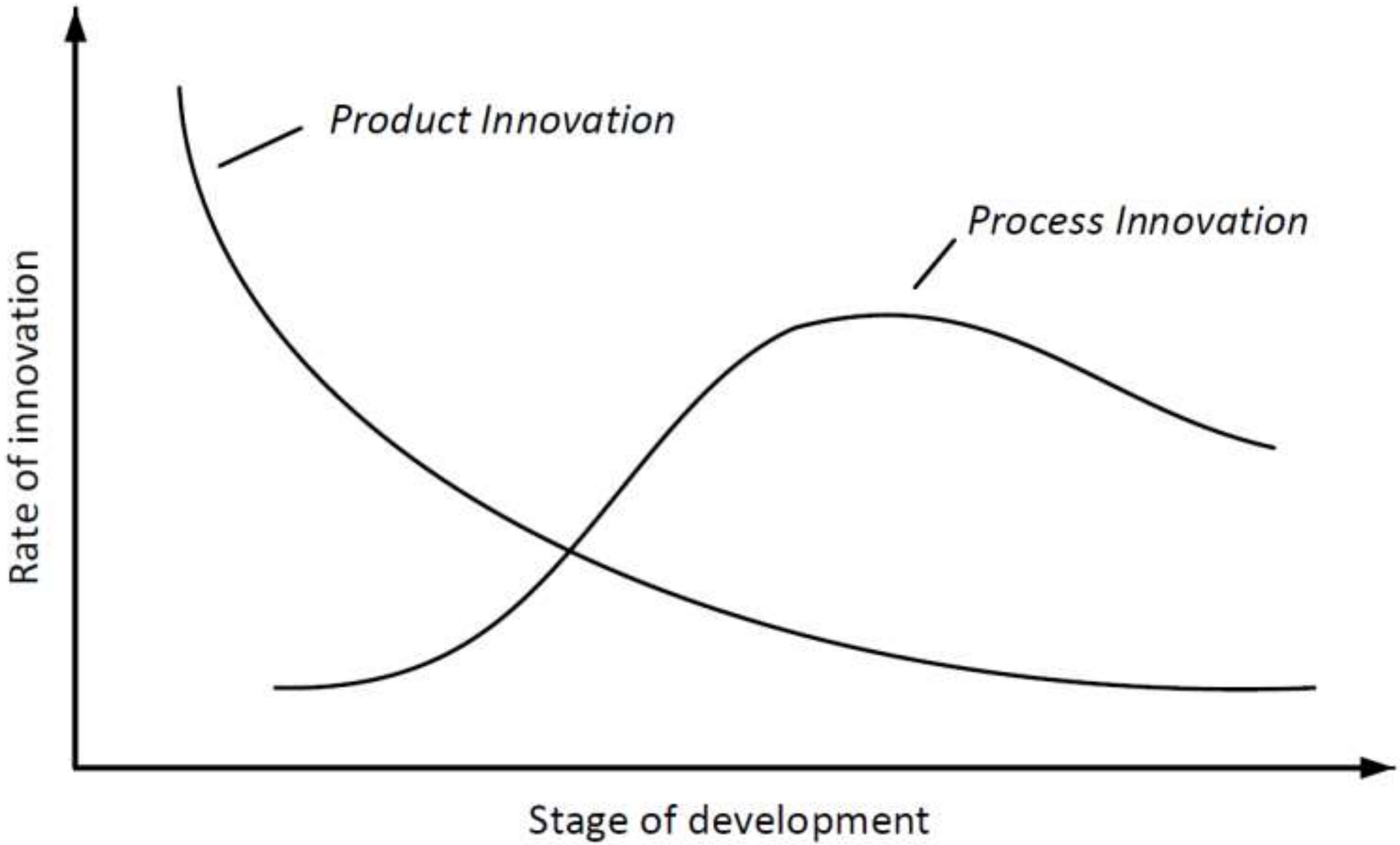
Positive motive:

- The urge of organizations to innovate

Reactive motive:

- Fight organizational entropy

The Abernaty-Utterback Model



Exercise 8.1: Organizational Entropy

- Can you identify business processes from your own experience that were efficient at some stage, but which have become unnecessarily complex?

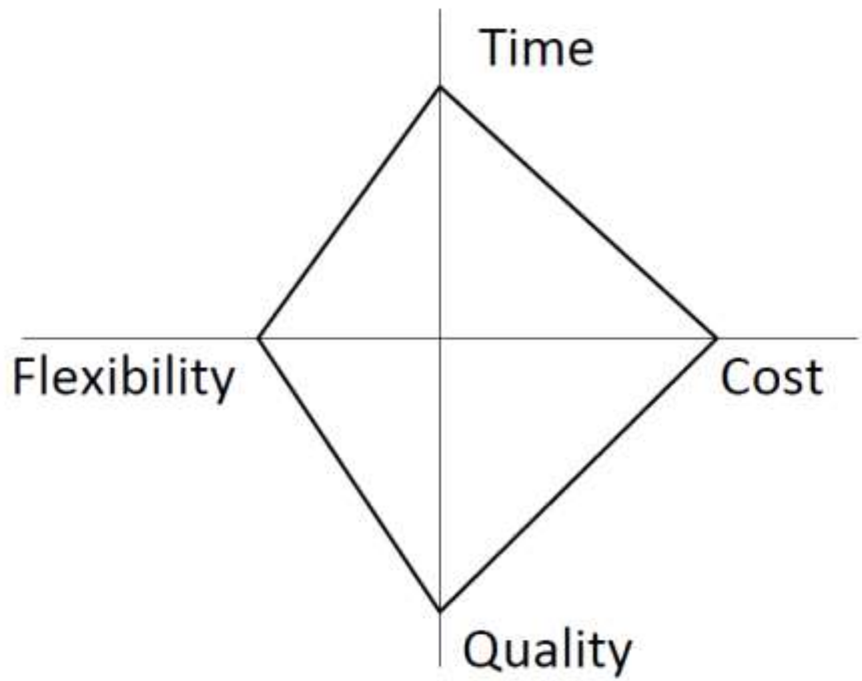
Elements of redesign

1. the internal or external *customers* of the business process,
2. the *business process operation* view, which relates to how a business process is implemented, specifically the number of activities that are identified in the process and the nature of each, and
3. the *business process behavior* view, which relates to the way a business process is executed, specifically the order in which activities are executed and how these are scheduled and assigned for execution,
4. the *organization* and the participants in the business process, captured at two levels:
   a) the organization structure (elements: roles, users, groups, departments, etc.), and
   b) the organization population (individuals: agents which can have activities assigned for execution and the relationships between them),
5. the *information* that the business process uses or creates,
6. the *technology* the business process uses, and
7. the *external environment* the process is situated in.

Process Redesign

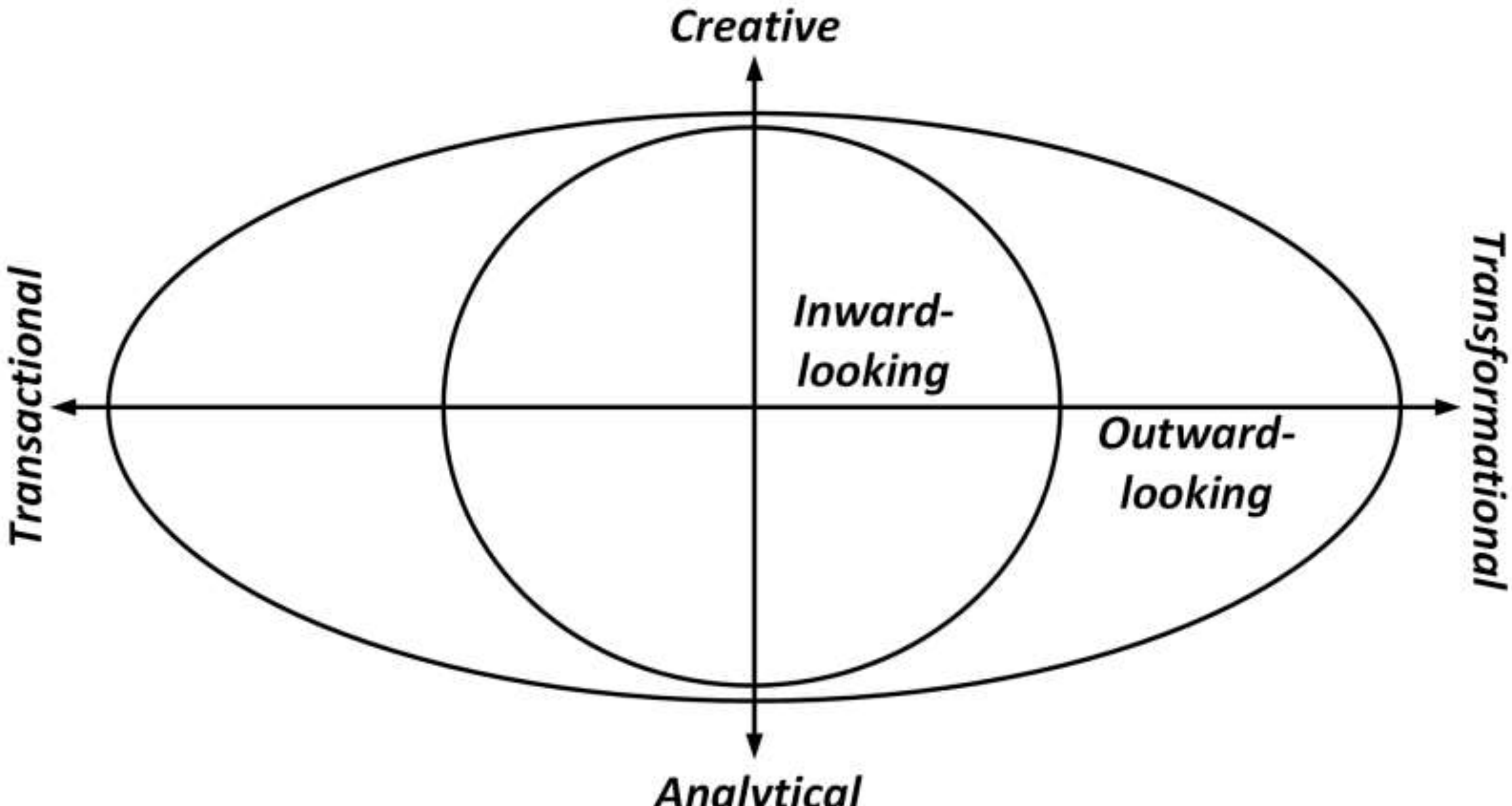
- Is a *substantial* and *intentional* change of a business process,
- Is primarily concerned with changing the *business process* itself, covering both its *operational* and *behavioral* view,
- Extends to changes that are on the interplay between on the one hand the process and on the other the *organization* or the *external environment* that the process operates in, the *information* and *technology* it employs, as well as the products it delivers to its *customers*.

The Devil's Quadrangle



"improving a process along one dimension may very well weaken its performance along another"

The Redesign Orbit



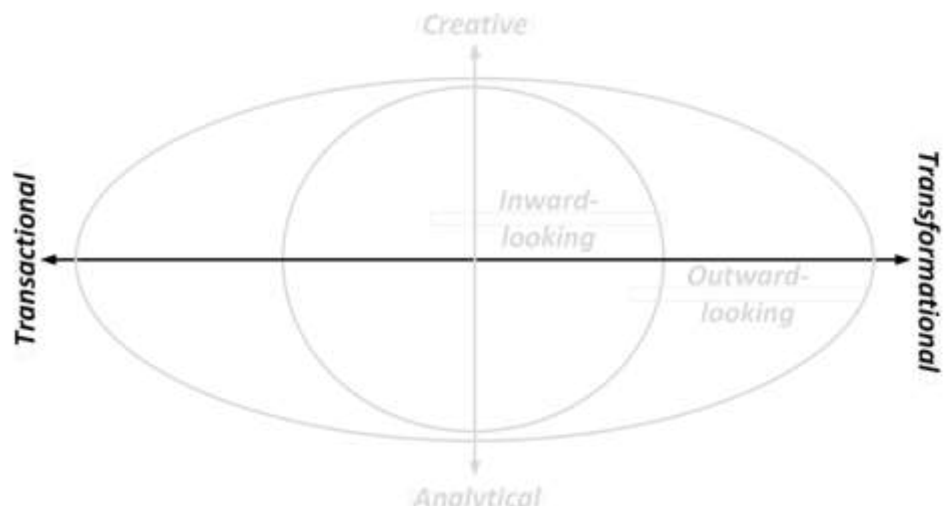
Ambition

Transactional Methods

- Seek to identify problems and resolve them incrementally, one step at a time
- Do not challenge the current process structure

Transformational Methods

- Aim to achieve breakthrough innovation
- Put into question the fundamental assumptions and principles of the existing process structure



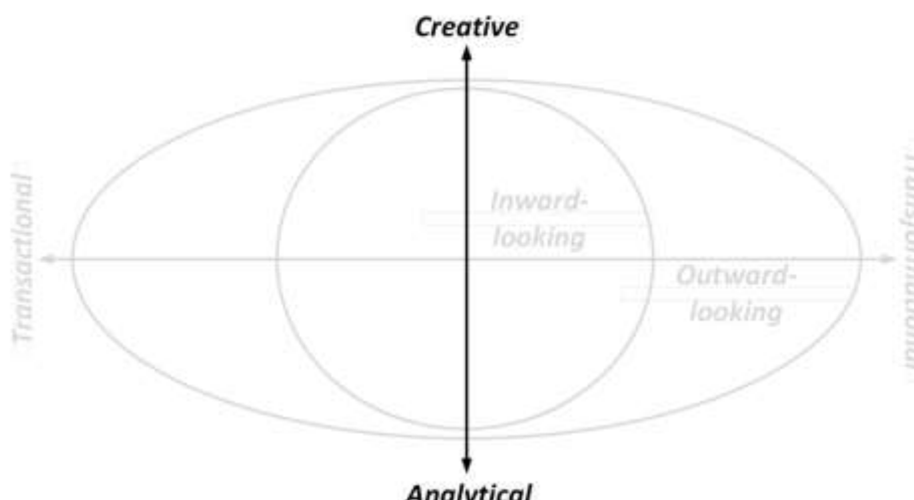
Nature

Analytical Methods

- Tend to have a strong mathematical and quantitative focus
- Embrace tools and technology

Creative Methods

- Rely on human creativity and ingenuity
- Embrace group dynamics



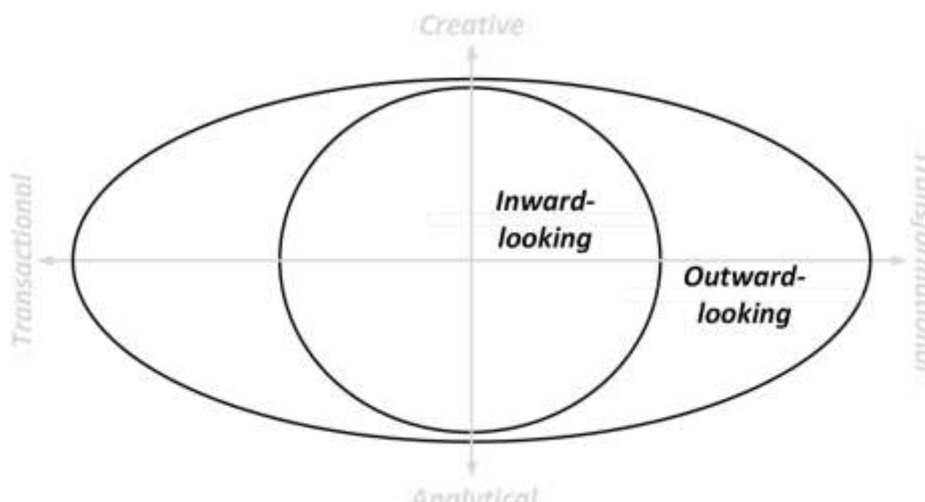
Perspective

Inward-looking Methods

- Consider the process from the perspective of the internal organization
- Draw from objectives and performance measurement

Outward-looking Methods

- Consider the process from an outsider's perspective
- Are driven by external opportunities and developments



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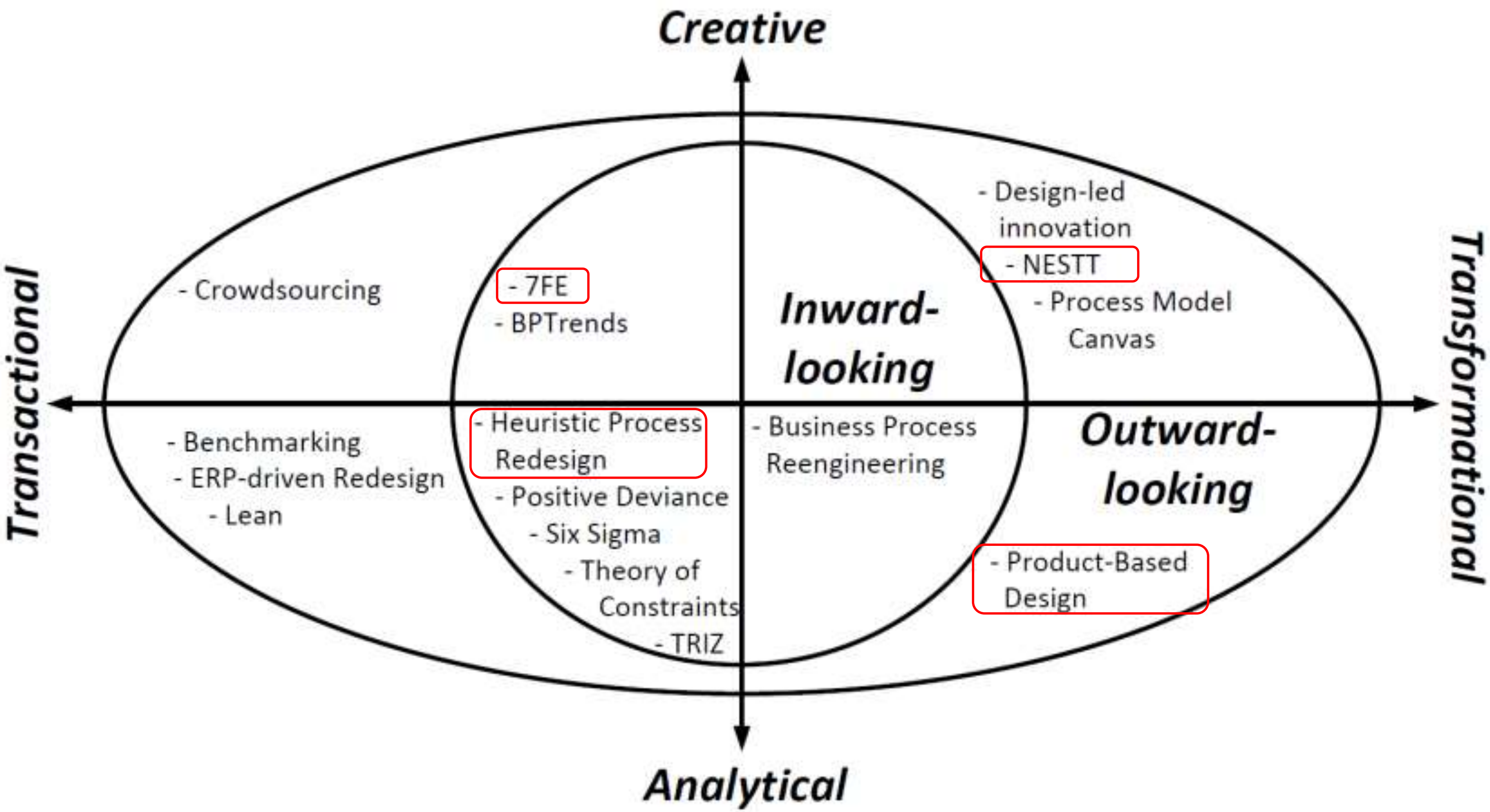
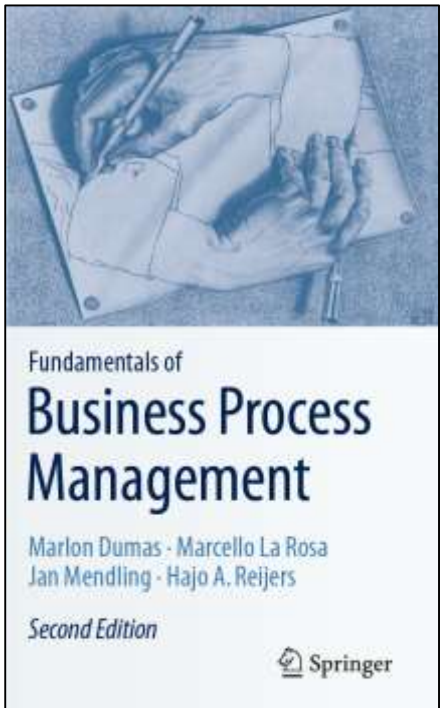


Chapter 8: Process Redesign



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1. The Essence of Process Redesign
2. Transactional Methods
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Transactional Methods

Analytical:
employ all kinds of tools, involve statistics, and are strongly rationalized

Inward-looking

- Six Sigma
- Theory of Constraints
- TRIZ
- Positive Deviance
- Heuristic Process Redesign

strongly focus on the existing process in an organization as a starting point

Outward-looking

- Benchmarking
- ERP-driven Redesign
- Lean

fundamentally outward-looking by taking an external perspective on process redesign

Transactional Methods

Creative:
unleashing the creativity of people

Inward-looking

- 7FE
- BPTrends

emphasis on engaging professionals that already play a role within a targeted business process

Outward-looking

- Crowdsourcing

tapping into the skills and knowledge of people outside their organizational borders

7FE

https://www.managementbyprocess.com/

7FE

Facilitators:

- Need to ask lots of “what if” and “why this” questions,
- Should not accept what he is or she is told (the first time),
- Must look for the second ‘right’ answer,
- Will regularly change the question and come it at from a different direction,
- Shall challenge the rules of the process,
- Need to rely on intuition.

Heuristic Process Redesign

- Uses a fixed list of redesign heuristics to determine potential improvement actions on an existing process.
- For each of the set performance goals, redesign team reflects on relevant heuristics that may be applied: A redesign heuristic is desirable to apply if it helps to attain the desired performance improvement of the process under consideration.
- Applicable and desirable redesign heuristics are clustered
- On basis of these clusters, a set of scenarios can be generated, each of which describes which redesign heuristics are applied in this scenario and, importantly, how this is done.

Time	Cost	Quality	Flexibility
Parallellism Case-based work	Activity elimination Empower	Empower Triage	Flexible assignment Centralization

Example Heuristics for Time

- **Parallelism**: “Put activities in parallel”. Activities in a business process are often ordered in a strictly sequential way even though there is no good reason for doing so. Some activities may well be carried out in an arbitrary order or even simultaneously. By allowing a less restrictive choice on the order in which activities are executed, a business process can be carried out faster.
- **Case-based work**: “Remove batch-processing and periodic activities”. Notable sources of delays in business processes exist where individual cases:
  - get piled up in a batch that is only processed once all its items are available, or
  - are slowed down by periodic triggers, e.g., a computer system is only available at one specific slot during the day.

  Getting rid of such constraints is in general a good way to significantly speed up a process.

Time	Cost	Quality	Flexibility
Parallellism Case-based work	Activity elimination Empower	Empower Triage	Flexible assignment Centralization

Example Heuristics for Cost

- **Activity elimination**: “Eliminate unnecessary activities”. Over time, processes get clogged up with activities that were useful at some point but have lost their purpose or rationale. Control activities, i.e., activities that are incorporated in a process to fix problems, are prime examples of non-value adding activities. Getting rid of unnecessary activities is an effective way to reduce the cost of handling a case.
- **Empower**: “Give workers decision-making authority”. In traditional settings, people have to authorize the outcomes of activities that have been performed by others. If workers are empowered to take decisions autonomously, this may render much of the work of middle managers superfluous, in this way reducing cost significantly.

Time	Cost	Quality	Flexibility
Parallellism Case-based work	Activity elimination Empower	Empower Triage	Flexible assignment Centralization

Example heuristics for Quality

- **Triage**: “Split an activity into alternative versions”. By creating alternative versions of an activity, it is possible to better deal with the variety of cases that need to be processed. An alternative activity essentially pursues the high-level goal of the original activity but is either geared specifically to a sub-category of cases that are being encountered (e.g., orders of special customers vs. all customers) or exploits the characteristics of the resource class that is assigned to it (e.g., senior clerks vs. all clerks). By aligning work more specifically to the properties of particular cases, the quality of work delivered improves.
- **Case assignment**: “Let participants perform as many steps as possible ”. If someone carries out an activity, then that person becomes acquainted at some level with the case for which the work is done. That knowledge accumulates with each activity that is done for the same case. By making one participant the preferred resource for any work that needs to be carried out for a particular case, this knowledge can be leveraged to deliver a high standard of work.

Time	Cost	Quality	Flexibility
Parallellism Case-based work	Activity elimination Empower	Empower Triage	Flexible assignment Centralization

Example Heuristics for Flexibility

- **Flexible assignment**: “Keep generic participants free for as long as possible”. Suppose that an activity can be executed by either of two available participants, then it should be assigned to the most specialized person. In this way, the likelihood to commit the free, more general participant to another work package is maximal. The advantage of this heuristic is that an organization stays flexible with respect to assigning work.
- **Centralization**: “Let geographically dispersed resources act as if they are centralized”. This heuristic is explicitly aimed at exploiting the benefits of a Business Process Management System (BPMS). After all, if a BPMS takes care of assigning work to process participants it becomes less relevant where these resources are located geographically. In this way, resources can be committed more flexibly.

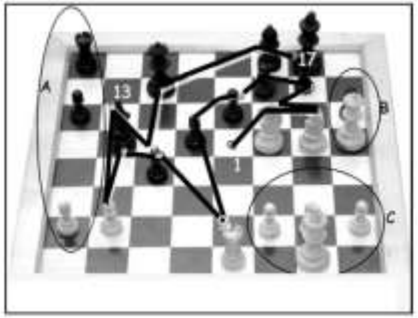
Time	Cost	Quality	Flexibility
Parallellism Case-based work	Activity elimination Empower	Empower Triage	Flexible assignment Centralization

Exercise 8.9: Negative Effects

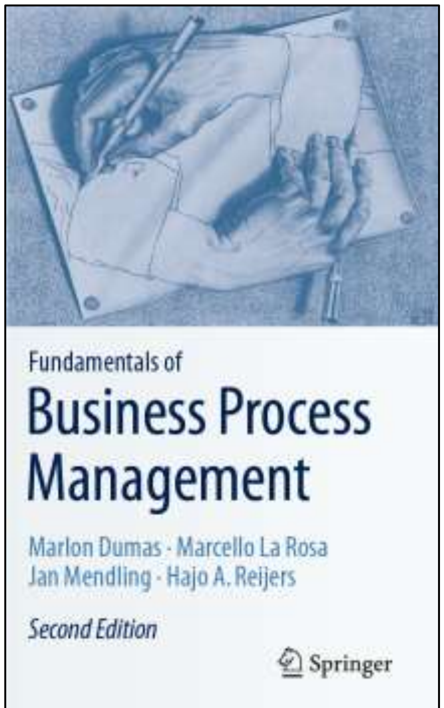
- In recognition of the Devil's Quadrangle, each heuristic can also have negative side-effects when applied. Can you imagine what negative impact the *Frequenz* redesign scenario may have on the performance of the rental car collection process in terms of Cost and Flexibility?

Full Set of Redesign Heuristics

- Control relocation
- Contact reduction
- Integration
- Case types
- Activity elimination
- Case-based work
- Triage
- Activity composition
- Resequencing
- Parallelism
- Knock-out
- Exception
- Case assignment
- Flexible assignment
- Centralization
- Split responsibilities
- Customer teams
- Numerical involvement
- Case manager
- Extra resources
- Specialist-generalist
- Empower
- Control addition
- Buffering
- Activity automation
- Integral technology
- Trusted party
- Outsourcing
- Interfacing



Chapter 8: Process Redesign



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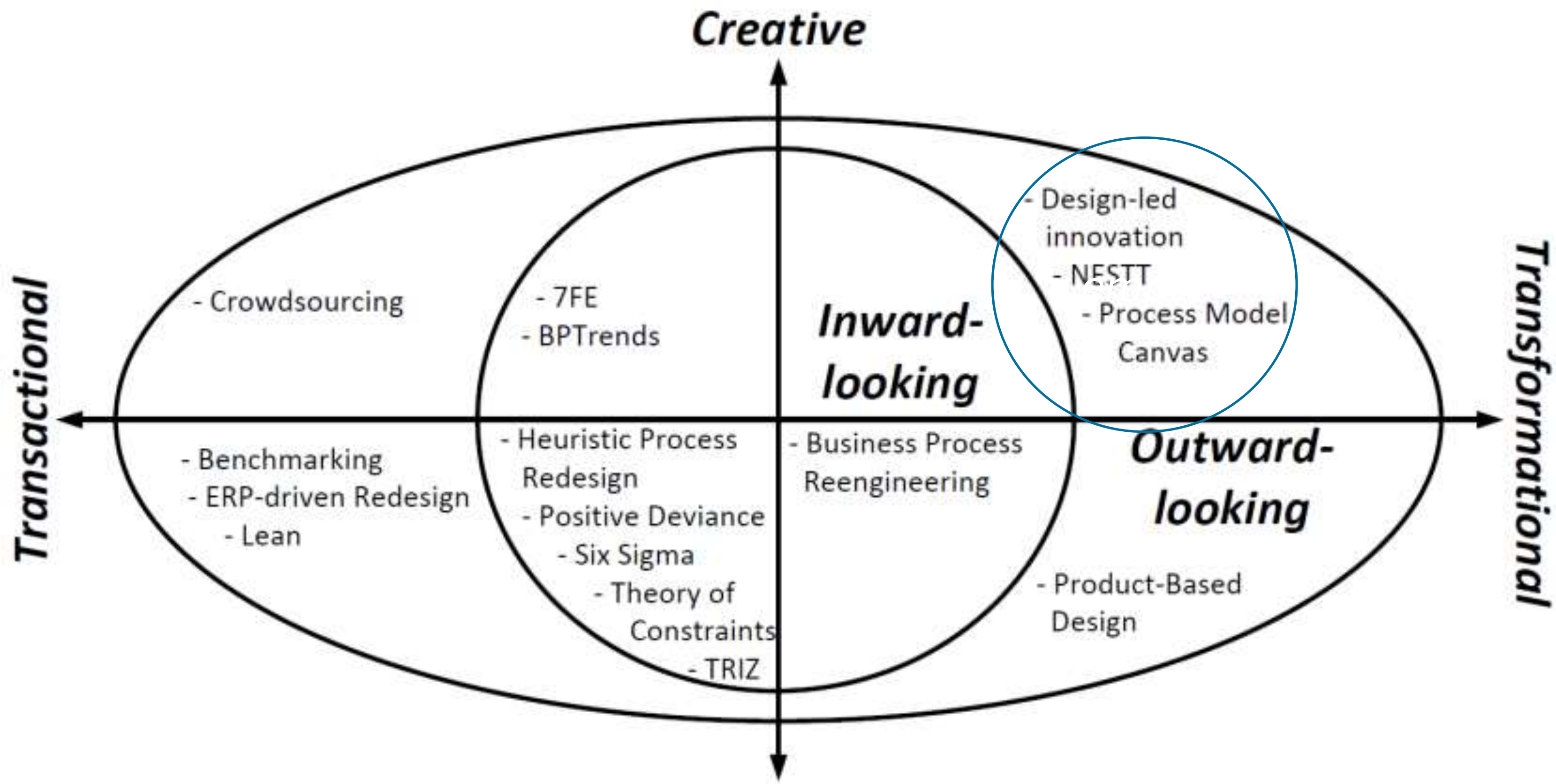


Transformational Methods

Creative: unleashing the creativity of people

Design-led Innovation:

- aims to provide organizations with an understanding of the deep emotional ties that consumers develop with their products.
- Its basic tenet is that people are not only served by the form and function of a product, but also through the experience its usage invokes.
- Based on this understanding, organizations may pursue innovations that customers do not expect, but which they eventually grow passionate about.

Transformational Methods

Creative: unleashing the creativity of people

NESTT: Navigate, Expand, Strengthen, and Tune/Take-off.

- Its defining feature is how participants in a workshop setting use the spatial affordances of a dedicated room.

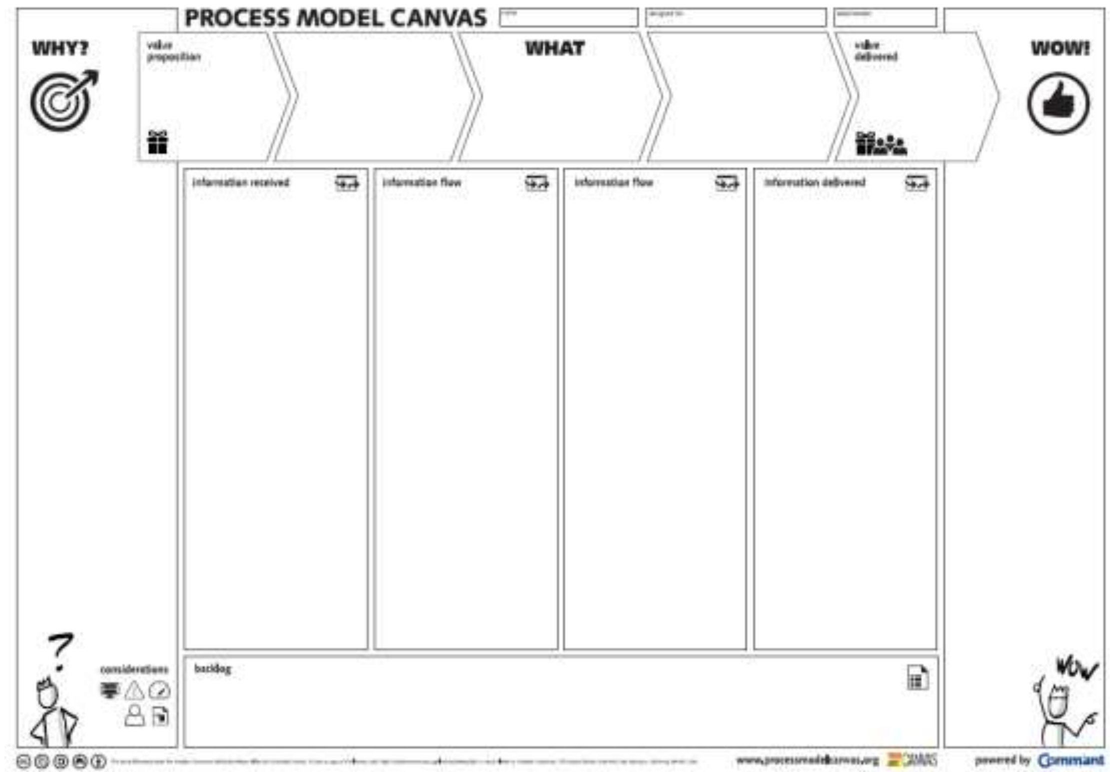
NESTT

Transformational Methods

Creative: unleashing the creativity of people

Process Model Canvas:

- allows firms to reason about the value proposition behind their business processes.



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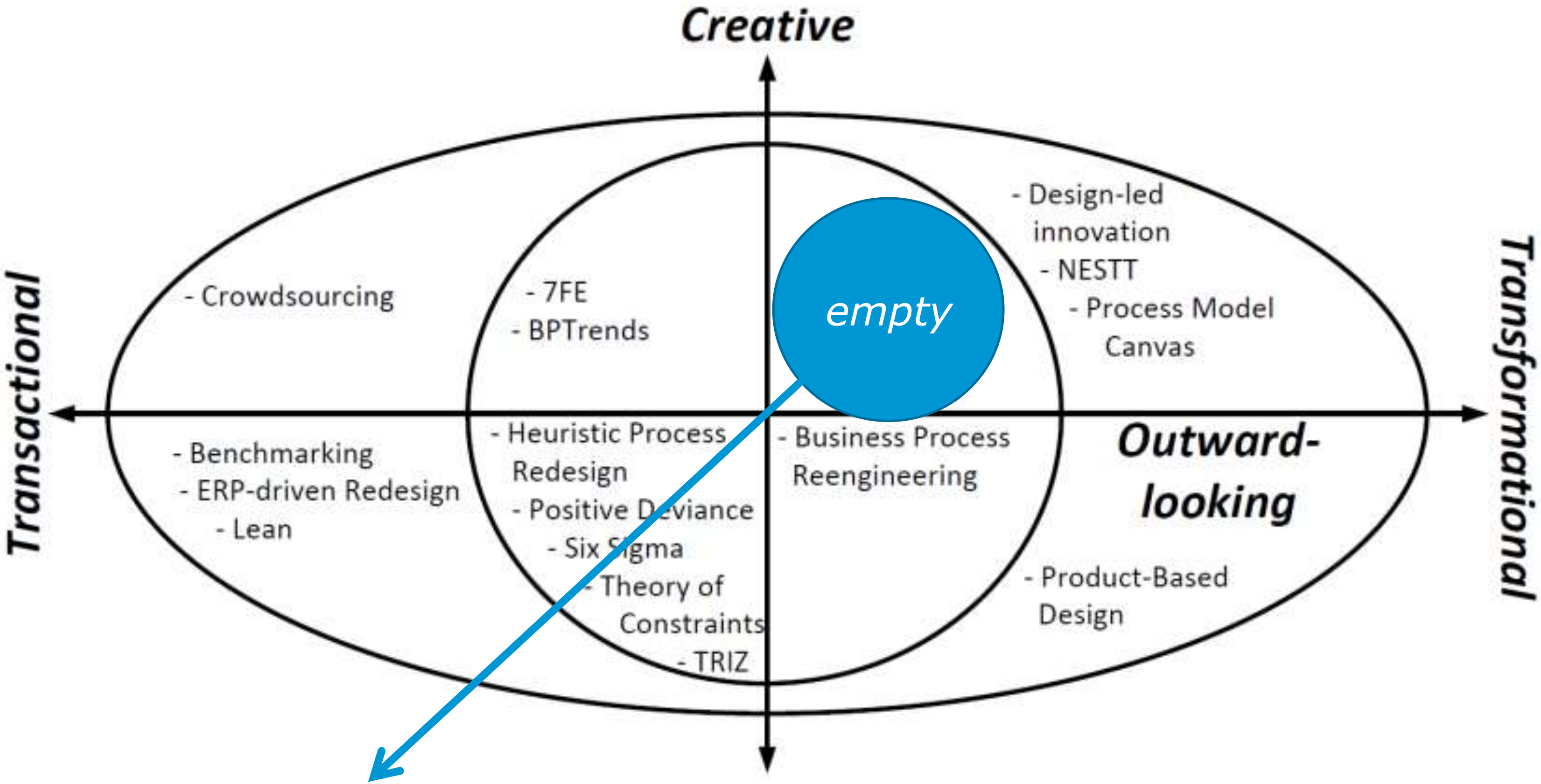
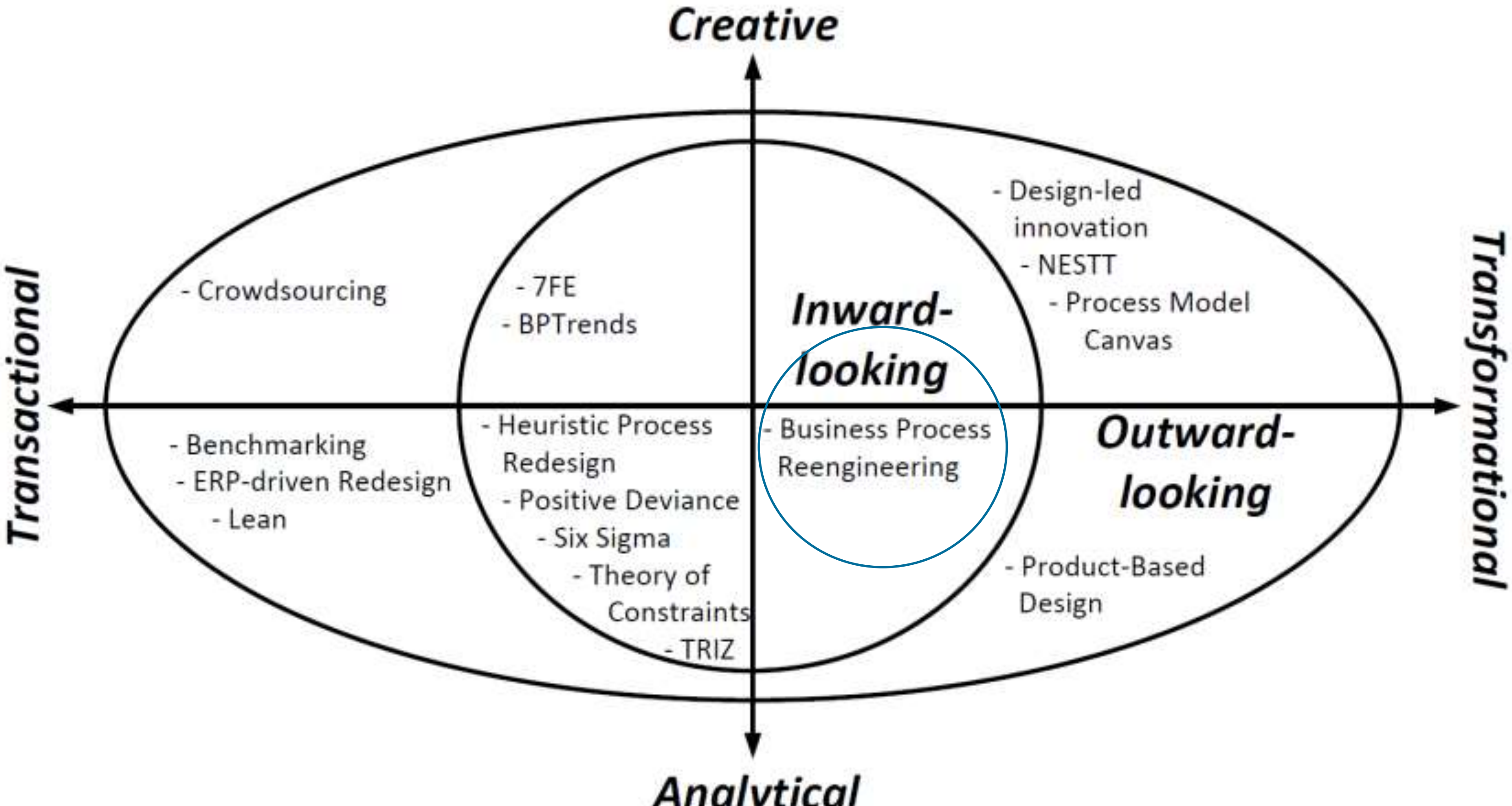


true transformations hardly emanate from reasoning from an internal perspective only

The Redesign Orbit



Transformational Methods

Business Process Reengineering

Michael Hammer collected the following insights:

1) No successful organization relies on piecemeal improvement of what was already carried out. Rather, strong ambition leads to huge rewards
2) While information technology is a crucial asset in redesigning business processes, it is necessary to go beyond pure automation of what is already being done
3) Organizations need to break away from a set of ingrained patterns of organizing work that prevent business processes from being carried out in an integrated, cross-functional way

Transformational Methods

Business Process Reengineering

Properties:

- The objective is to completely overhaul a process, which puts it in the **transformative** sphere.

- It is **analytical** because it relies on such a set of clearly defined principles, in contrast to what a group of people comes up with,

- It is mostly **inward-looking** since it still operates within the scope and context of the existing process it aims to overhaul.

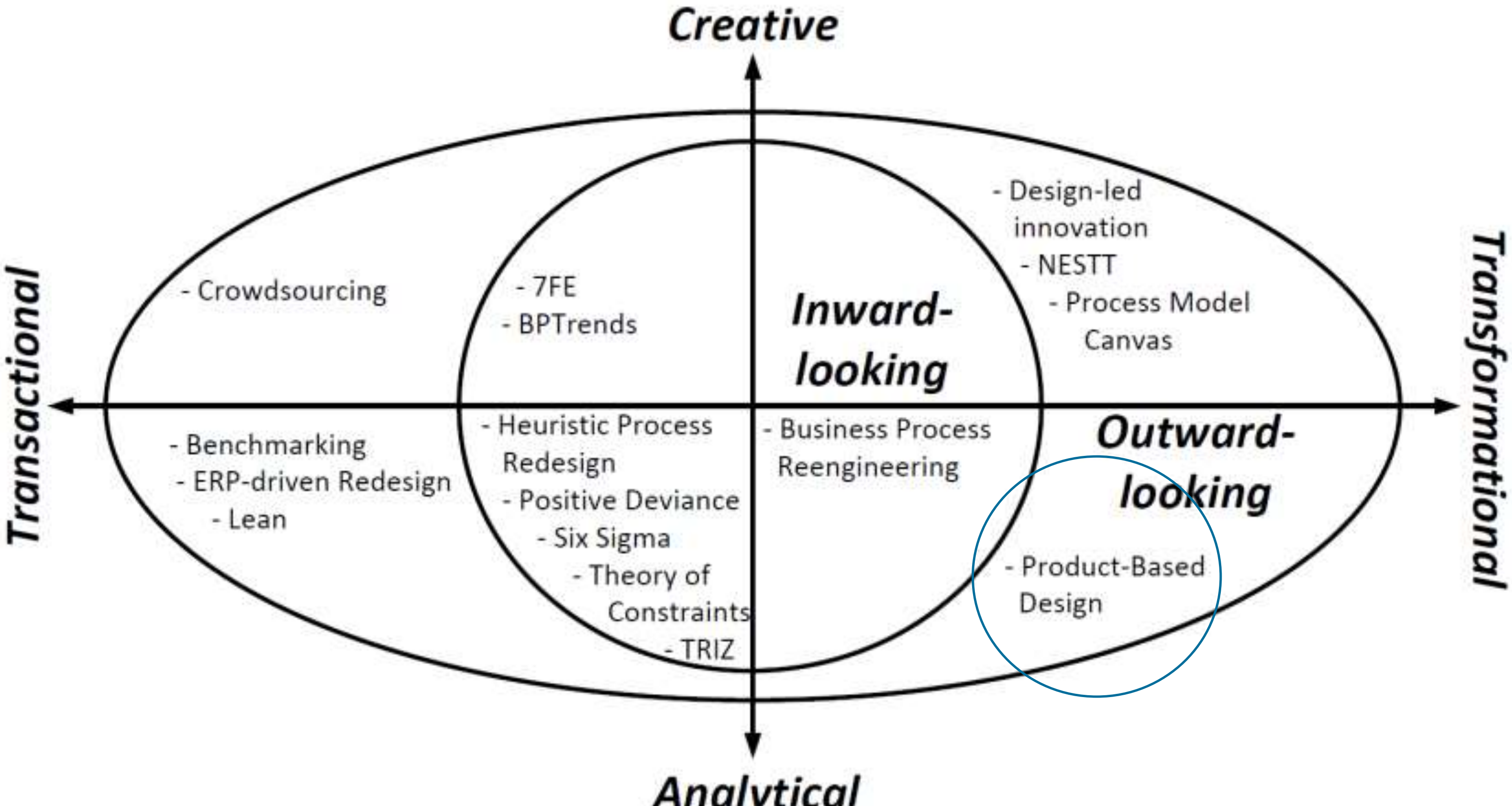
Transformational Methods

Business Process Reengineering

Redesign principles:

1. Make sure that information is captured fresh, at the moment it is produced, and at the source by the stakeholder who is producing it..
2. Integrate information processing work, i.e., work that involves capturing or processing information, with the real work where this information is produced.
3. Let those who have an interest in the output of a process not only participate in it but drive it all the way.
4. Put every decision point in a process preferably at the place where work is performed.

The Redesign Orbit



Transformational Methods

Product-Based Design

Properties:

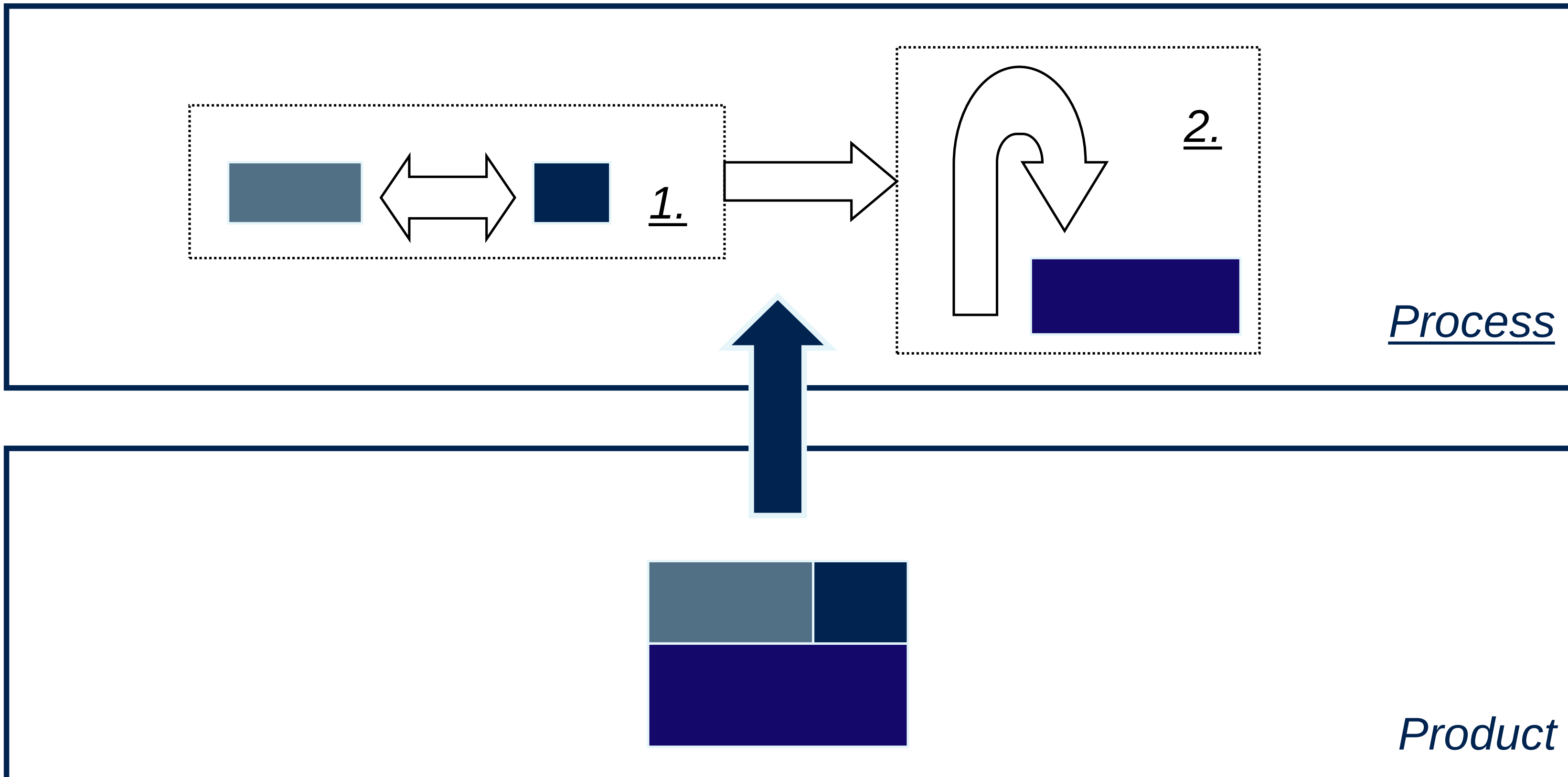
- The objective is to completely overhaul a process, which puts it in the **transformative** sphere.

- It is **analytical** in nature since it relies on a formal, almost purely algorithmic way of developing a new business process.

- It is **outward-looking** because of the artifact that takes center stage in this method: It is the product that a business process aims to deliver.

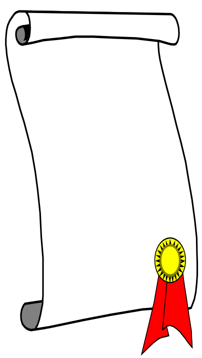
Transformational Methods

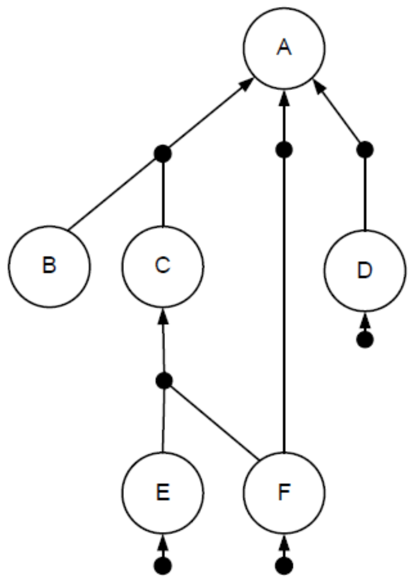
Product-Based Design

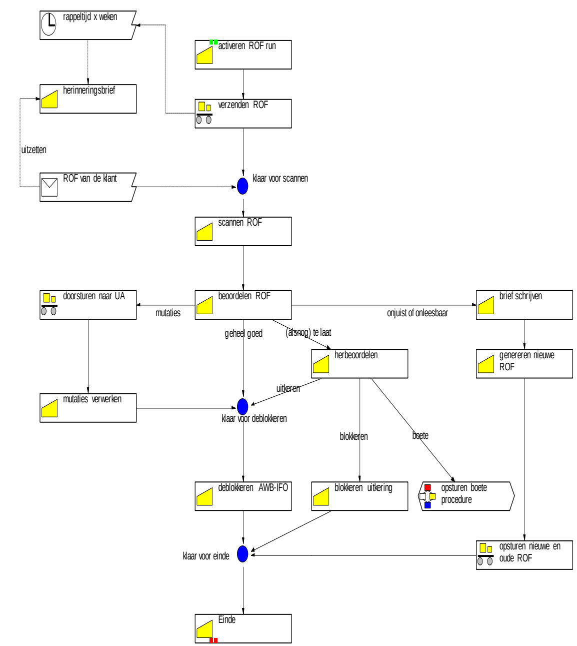


Transformational Methods

Product-Based Design

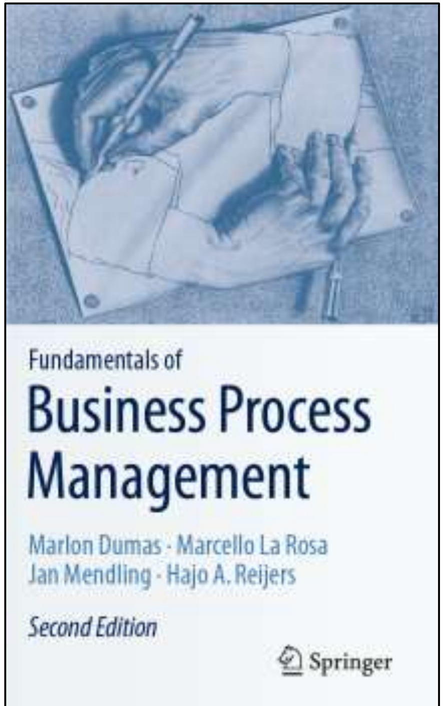






Product specifications → **Product Data Model** → **Process**

Chapter 8: Process Redesign



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Recap

- Process redesign can be motivated from a positive and a reactive angle
- Process redesign can be understood by considering the different elements of a process:
  - customers, business process operation, business process behavior, organization structure, organization population, information, technology, and the external
- The Devil's Quadrangle clarifies that many redesign options have to be discussed from the perspective of a trade-off between time, cost, quality, and flexibility
- The Redesign Orbit provides a spectrum of redesign methods using three axes:
  - nature, ambition, and perspective
- Redesign methods have their own distinctive characteristics

MIDTERM & FINAL EXAMINATIONS

Midterms (2 HOURS, 1:30-3:30)

- Process Discovery and
- BPMN
- Use of the following:
  - Events
  - Sequence Flow
  - Activities
  - Gateways (XOR, OR, AND)
  - Pools and Lanes
  - Message Flow
  - Business Objects

- Finals (4 HOURS, 1:30-5:30)
- Process Analysis (3 HOURS, 1:30-4:30)
  - Qualitative
    - Issue Register
    - Why Why Diagram
    - Fishbone
  - Quantitative
    - Issue Register - Cost
    - Flow Analysis (Cycle Time
- Process Redesign (10minutes)
  - Pitch Deck
    - Present Improvement Proposal